

Comments Submitted on Proposed Revisions to the NERC Rules of Procedure

NERC Rules of Procedure Appendix 2 (Definitions Used in the Rules of Procedure); Appendix 5A (Organization Registration and Certification Manual); and Appendix 5B (Statement of Compliance Registry Criteria)

NERC posted proposed revisions to its Rules of Procedure to develop a new registered entity type, called Computational Load Entity, for comment from April 1, 2026 to May 15, 2026. The entities in the following table responded during the comment period, either submitting comments or indicating support of other entities' comments. This pdf package includes the comments received during this posting period, with personal contact information removed.

List of Responding Entities

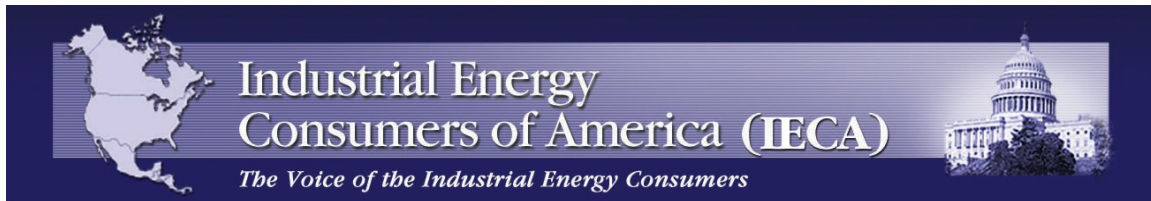
Entity	
1	Industrial Energy Consumers of America
2	Microsoft
3	ISO/RTO Council Standards Review Committee
4	Electricity Customer Alliance, Electricity Consumers Resource Council, and Industrial Energy Consumers of America
5	American Electric Power
6	Alphataraxia Management LP
7	Ameren*
8	ATC*+
9	AZPS*+
10	City of Santa Clara (dba Silicon Valley Power)*+
11	Southwest Power Pool
12	Equinix
13	Dominion Energy*
14	Corscale Data Centers**
15	AIONTRACKS, Inc.
16	Crusoe Technologies LLC
17	DC Energy Management, LP
18	Digital Power Network

19	Electricity Consumers Resource Council
20	Entergy
21	Arkansas Electric Cooperative Corporation
22	Edison Electric Institute, American Public Power Association, Large Public Power Council, National Rural Electric Cooperative Association, Transmission Access Policy Study Group, and National Association of Regulatory Utility Commissioners (Trade Associations)
23	FirstEnergy*
24	Google**
25	Independent Electricity System Operator
26	Invenergy
27	Midcontinent Independent System Operator
28	Data Center Coalition
29	Bonneville Power Administration
30	Arevon Energy, Inc.
31	New York Power Authority*
32	Oracle**
33	Persistence Analytics Group LLC
34	PJM Interconnection
35	Eversource*+
36	CenterPoint Energy Houston Electric, LLC*
37	Consumers Energy
38	PacifiCorp
39	Southern Company*
40	Solar Energy Industries Association
41	Sacramento Municipal Utility District and Balancing Authority of Northern California*
42	Salt River Project*
43	American Chemistry Council

*supports Trade Associations

**supports Data Center Coalition

+means support and additional comments



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May 15, 2026

North American Electric Reliability Corporation
1401 H Street NW, Suite 410
Washington, DC 20005
ROPcomments@nerc.net

Re: Proposed Computational Load Entity Registry

On behalf of the Industrial Energy Consumers of America (IECA), we urge you not to move forward with the proposed Computational Load Entity Registry. Manufacturing self-generation units (CHP/WHP) operate as stable, predictable resources that do not present the reliability risks this proposal is intended to address. There is no evidence to suggest that these units are a threat to the grid.

The proposed rule is intended to address reliability gaps associated with emerging computational loads but does not identify a demonstrated reliability gap for CHP/WHP resources. We are not data centers. We do not operate like data centers and we should not be treated like them under contemplated regulations.

The proposed criteria appropriately focus on large, high-density computational loads comprised of information technology equipment with distinctive and less predictable electrical behavior, which is not characteristic of CHP/WHP facilities driven by thermal and industrial processes. Unlike computational loads, which can fluctuate rapidly, CHP/WHP units are thermal-following and operate with stable, continuous output that supports grid stability. Industrial facilities typically fall within conventional load profiles and may only include de minimis levels of information technology equipment, consistent with NERC's distinction based on differing electrical behavior. The proposed changes unnecessarily increase our costs and economic and regulatory uncertainty.

One hundred percent of IECA's members are manufacturing companies from every industry. Many of these companies have been generating their own power and steam since the late 1970s, using CHP and/or heat to power systems.

According to EIA there are over 900 CHP units across the U.S. that have a rated capacity of about 60,000 MW. They have operated successfully day and night without any negative impacts to grid reliability. Some of them provide valuable balancing benefits to the grid by producing less at night and buying power from the grid when demand by other sectors are low. This predictable behavior supports grid balancing and supports frequency stability through continuous operation that differs fundamentally from the volatility of data center loads.

Importantly, some of these units have a history of being called upon to provide power in times of power shortages, thereby supporting grid reliability.

Existing NERC reliability standards and interconnection requirements, including FAC-001 and FAC-002, already provide appropriate oversight and study requirements for industrial facilities and their interaction with the Bulk Power System. No additional regulatory gap has been identified that would justify expanding registration to CHP/WHP under a new functional entity classification.

We are a resource and we look forward to working with you to help ensure any new registry is appropriately scoped to address emerging, high-variability computational loads while avoiding unintended inclusion of stable, conventional industrial generation.

Sincerely,

Paul N. Cicio

Paul N. Cicio

President & CEO

The Industrial Energy Consumers of America is a nonpartisan association of leading manufacturing companies with \$1.3 trillion in annual sales, over 12,000 facilities nationwide, and with more than 1.9 million employees. One hundred percent of IECA members are manufacturing companies whose competitiveness is largely determined by the cost and reliability of natural gas and electricity. IECA's sole mission is to reduce and avoid energy costs and increase energy reliability through advocacy in Congress and regulatory agencies, such as the Federal Energy Regulatory Commission (FERC). IECA membership represents a diverse set of industries including chemicals, plastics, steel, iron ore, aluminum, paper, food processing, fertilizer, insulation, glass, industrial gases, pharmaceutical, consumer goods, building products, automotive, independent oil refining, and cement.



ELECTRICITY CUSTOMER ALLIANCE

To: North American Electricity Reliability Council
ROPComments@nerc.com

Re: Comments on Proposed Computational Load Entity Rules of Procedure Revisions

The Electricity Customer Alliance (ECA), on behalf of itself and the undersigned customer groups, is pleased to submit comments on NERC's proposed changes to its Rules of Procedure to create a new functional entity type, called "Computational Load Entity", and define the criteria that will require such entities to register with NERC and comply with current or future mandatory and enforceable Reliability Standards. ECA appreciates the opportunity to submit these comments and the efforts that NERC and its staff have made to engage with customers through the course of its analysis of the impacts of emerging large loads on bulk power system reliability. **For the reasons discussed below, ECA asks that NERC decline to move forward with the proposal at this time and instead conduct further examination of the most cost-effective and efficient means of addressing the reliability risks it has identified with respect to emerging large loads.**

ECA is a growing national coalition dedicated to advancing customer-centric solutions that modernize the grid, support economic growth and the development of digital infrastructure and advanced manufacturing, and keep electricity affordable and reliable for all users. Members and participants in ECA include a diverse range of customers and customer groups, from large customers in the technology, industrial, and manufacturing segments, to smaller commercial customers and residential ratepayer advocates. ECA works to elevate customer perspectives and coalesce and align customer segments to identify and advance common ground approaches to the challenges of load growth, grid expansion and modernization, electricity market governance and accountability, and rising electricity affordability concerns.

Creating the proposed new functional entity type, "Computational Load Entity," would mark a dramatic departure from historic practice by, for the first time, subjecting end use customers to mandatory and enforceable reliability standards. To date, the mandatory and enforceable reliability standards framework has focused on the planning and operational activities of entities that generate, transmit, or deliver electricity to end use customers to ensure that these activities support reliable operation of the bulk power system. This includes, among other things, planning, operational practices, communications, training, and system protection standards that these entities must individually and collectively perform to ensure reliable operations in accordance with the requirements of Section 215 of the Federal Power Act. In short, the existing reliability standards framework is focused on the entities that have operational control of the bulk power system and that regularly engage in coordinated planning and communications protocols to ensure reliable operations.



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NERC has not demonstrated why bringing end use customers into this framework for the first time would be the most effective and efficient means of solving the emerging reliability challenges it has identified. NERC's Large Load Working Group (LLWG) has found that emerging large loads (which it proposes to define as "Computational Loads" like data centers or cryptocurrency operations) may pose certain new risks to reliable operations, including operational risks from sudden load drops, ramping characteristics, and inability to ride through disturbances. In addition, the LLWG identified gaps in existing reliability standards and requirements that do not account for the unique characteristics of these emerging loads, such as planning and interconnection obligations and modeling and study requirements. With respect to large load interconnection processes, the LLWG identified the lack of consistent interconnection requirements among current registered entities to ensure the identification of potential impacts to the bulk power system, such as requirements related to voltage/frequency disturbance ride-through, sensitive protection settings, ramping, power quality, and oscillation. The LLTF also noted gaps in the assignment of responsibility among existing registered entities for assessing and mitigating these potential risks.

NERC has not fully explained why an unprecedented expansion of registration requirements to end use customers is the most effective strategy for filling these gaps in requirements and assignments of responsibility. The LLTF's gap assessment's first recommendation notes that "[t]here are multiple high-impact risks to the [bulk power system] from large loads that NERC registered entities cannot adequately address," a statement followed by a stark conclusion that NERC should "pursue registration of a type of entity (or types of entities) that is able to perform specific functions to address the risks."¹ NERC repeats this conclusion in the "Technical Reasoning" document accompanying the proposed revisions to the Rules of Operating Procedure, stating that "[d]ue to the number of existing and anticipated large loads needing power to support computational processes and the unforeseen and difficult-to-explain dynamic electrical behavior of these users of the [bulk power system] as identified by the LLWG, NERC determined that it was appropriate to propose a new functional entity type that focuses on loads with these characteristics."²

These broad statements do not answer the fundamental question of why existing NERC registered entities could not adequately address these impacts with changes to the existing reliability standards, or the development of new reliability standards, to fill the information, modeling, study, and operational coordination gaps that have been identified. The two Large Load Task Force (LLTF) white papers that form the basis for the recommendation to create a new functional entity, and the

¹ NERC, "Assessment of Gaps in Existing Practices, Requirements, and Reliability Standards for Emerging Large Loads: NERC Large Loads Working Group White Paper" at 53 (March 2026), *available at* <https://www.nerc.com/globalassets/our-work/guidelines/reliability/white-paper---assessment-of-gaps.pdf>.

² NERC, "Technical Reasoning: Proposed Computational Load Entity" at 2 (April 2, 2026), *available at* <https://www.nerc.com/globalassets/who-we-are/rules-of-procedure/proposed/cle-technical-reasoning-april-2026-posting.pdf>.



ELECTRICITY CUSTOMER ALLIANCE

other available evidence, including the two industry alerts NERC has issued urging existing registered entities to take additional steps to address large load impacts, likewise do not answer this question.

This fundamental question should be answered before taking the unprecedented step of expanding the universe of registered entities beyond those with direct operational control of the bulk power system. Adding end use customers to the NERC registry and the framework of mandatory and enforceable reliability standards would add new complexities to the compliance regime and could create unintended consequences for that regime. For example, end use customers do not have the same operational control over the bulk power system or visibility into operations that existing registered entities possess, nor are they integrated into the communications and protocols that have long existed between these entities. These realities will likely create added complications for end use customers in complying implementing operational controls in real-time when compared to existing registered entities. In addition, integrating an entirely new set of registered entities for the first time could result in new process or analysis gaps or require new modes of communication, requiring additional work that could detract from or stall progress in addressing the emergent reliability concerns identified by the LLWG. Moreover, deflecting a subset of reliability obligations to end use customers instead of the entities with direct bulk power system operational control could fragment and weaken existing accountability structures.

Reforming the existing reliability standards or developing new standards to fill the information, modeling, study, and operational coordination gaps that have been identified would not mean that emerging large loads are free of any responsibility for contributing to the reliable operation of the bulk power system. Load interconnection requirements and communication protocols implemented by registered entities would place appropriate obligations on those large load customers while retaining the focus of the NERC standards framework on those entities who have direct operational control of the bulk power system. The difficulty of defining “Computational Loads” and the fact that customer loads are not a monolith and have different operational profiles and potential grid impacts, both of which NERC has acknowledged, makes it all the more appropriate to focus on interconnection requirements implemented through existing registered entities rather than expanding the registered entities that must be involved.

In addition, before taking such a significant step to expand the universe of registered entities, NERC should solicit the views of the Federal Energy Regulatory Commission (FERC), states, and other key policymakers, and engage them in further examination of the most effective approach to address the identified reliability risks. FERC is ultimately responsible for implementation of the mandatory reliability standards framework envisioned by Congress in Section 215 of the Federal Power Act, including determining the jurisdictional reach of Section 215 to “users, owners and operators of the bulk-power system.” NERC’s proposed criteria directly implicate jurisdictional questions, particularly its proposal to apply registration requirements based on aggregated load at a single



ELECTRICITY CUSTOMER ALLIANCE

point of interconnection at a voltage of 60 kV or higher, which is above the 100 kV threshold for determination of bulk power system facilities. FERC guidance on these questions before taking additional steps to extend the NERC compliance framework well beyond where it has been historically, and beginning the process of placing the obligations of that framework on end-use customers, is critical. Moreover, other federal and state policymakers likely have views on the efficacy of such a change and how it could impact consumer costs. Additional process (perhaps convened by FERC) is important to transparently address these issues before moving forward.

Finally, while NERC has identified important reliability considerations of emerging large loads that require immediate attention, it is not clear that expanding the universe of registered entities is necessary on the aggressive schedule NERC is pursuing. We understand that the initial reliability standards developed under the draft Standards Authorization Request issued in April will not apply to the proposed new Computational Load Entities. In addition, the Level 2 and 3 Alerts that NERC has issued in the past several months have identified important steps that existing registered entities should take now to address emergent reliability concerns identified by the LLWG. As a result, there is ample time to engage in additional examination and fact finding before finalizing a decision on registration of Computational Load Entities.

For these reasons, we urge NERC to reconsider adding additional registered entities at this time and instead examine whether reforming existing standards and the responsibilities placed on existing registered entities would be a more effective and efficient approach to address the identified gaps. We look forward to further engaging with NERC and stakeholders on these important reliability issues.

Sincerely,

Electricity Customer Alliance (ECA)
Electricity Consumers Resource Council (ELCON)
Industrial Energy Consumers of America (IECA)

American Electric Power Comments on Proposed Revisions to NERC Rules of Procedure for Computational Load

American Electric Power (AEP) appreciates the opportunity to comment on the proposed revisions to the North American Electric Reliability Corporation (NERC) Rules of Procedure (ROP) related to Computational Load. AEP supports NERC's objective of protecting the Bulk Power System (BPS) from reliability risks associated with large Computational Loads and understands the need to develop registration criteria for owners and operators of Computational Load.

AEP offers the following comments and recommended modifications to help ensure the proposed changes are clear, consistent with existing NERC registration models, and appropriately focused on Computational Loads that pose a material risk to BPS reliability.

AEP Proposes the Following Modifications:

Appendix 2 (Definitions Used in the Rules of Procedure)

1. Addition of New Definition Computational Load Facility (CLF)

AEP supports the Trade Associations' recommendation to add a new definition Computational Load Facility (CLF):

"Computational Load Facility" means a collection of electrical equipment located at one or more buildings, structures, or installations that is designed to receive power, has an aggregate connected maximum peak demand of Computational Load greater than or equal to 75 MW, and is served through electric facilities at a voltage level greater than or equal to 100 kV from one or more points of connection.

A clear definition of a Computational Load *Facility* is essential to the success of Project 2026-02. Existing Standards that address similar types of essential actions being proposed for data centers but are applicable to other comparable equipment (like generators) consistently rely on a *Facility*-type definition. This is consistent with how Generator Owner/Operator and Generating Facility are treated, defined, and included in the registration criteria.

2. AEP agrees with the Trade Associations' recommendation to replace CL Entity (CLE) with CL Operator (CLOP) and CL Owner (CLO):

"Computational Load Owner" means the end-user entity that owns and maintains a Computational Load Facility, **and/or** the entity that owns and maintains a Computational Load Facility that hosts end-users that receive electric power **at a Computational Load Facility**.

“Computational Load Operator” means the end-user entity that operates a Computational Load Facility, **and/or** the entity that operates a Computational Load Facility that hosts end-users that receive electric power **at a Computational Load Facility**.

AEP recommends the same “Owner” and “Operator” Model for Computational Loads to remain consistent with the registration criteria currently in the NERC Rules of Procedure for Transmission Owner (TO), Transmission Operator (TOP), Generator Owner (GO), and Generator Operator (GOP). This structure is in place because for Transmission and Generation facilities, there are often situations that occur where the owner of the facilities is different from the operator of the facilities. This registration criteria structure allows the different entities to be responsible only for the activities that they are contractually set up for as the owner and/or operator.

3. AEP supports the Trade Associations’ recommendation to clarify the Computational Load (CL) definition:

“Computational Load” means Load comprised **primarily** of electric power demand from **power electronic-based** information technology equipment.

Appendix 5A (Organization Registration and Certification Manual)

4. As discussed in Comment 2 above, AEP agrees with the Trade Associations’ recommendation to replace CL Entity (CLE) with CL Operator (CLOP) and CL Owner (CLO) to align with existing NERC Owner/Operator registration frameworks.

Appendix 5B (Statement of Compliance Registry Criteria)

5. AEP supports the Trade Associations’ recommendation to either (a) increase the proposed Computational Load thresholds or (b) pursue an approach comparable to the Category 1/Category 2 registration structure currently used for Generation.

a. Increase Computational Load Threshold

AEP recommends that NERC increase the proposed thresholds from 20 MW and 60 kV to 75 MW and 100 kV for Computational Load Facilities and eliminate the 1MW Computational Load criterion.

The proposed thresholds are set unnecessarily low and would result in the registration of a substantial number of facilities that do not pose a material risk to the reliability of the Bulk Power System (BPS) and, thus, do not justify the burden of NERC registration. The current Rules of Procedure, Registration Criteria, and Standard Drafting efforts related to Computational Loads should focus on those loads that pose the most meaningful risk to the BPS.

Accordingly, AEP recommends establishing a threshold of greater than or equal to 75 MW connected at greater than or equal to 100 kV for Computational Load Facilities and elimination the 1MW Computational Load criterion.

b. Category 1/Category 2 Registration Approach

NERC should pursue a Category 1/Category 2 approach for Computational Loads.

If a threshold lower than 75 MW is deemed necessary, AEP recommends that NERC enhance the Technical Reasoning document to reflect the analysis and supporting evidence demonstrating the need for the lower facility threshold for BPS reliability. In addition, AEP recommends that NERC adopt the same registration criteria model that was developed for Inverter-Based Resources (IBRs) with the Category 1 GO/GOP and establish a new Category 1 and Category 2 structure with the proposed definitions:

- **“Computational Load Facility”** means a collection of electrical equipment located at one or more buildings, structures, or installations that is designed to receive power, is served through electric facilities from one or more points of connection **and meets the criteria in either Category 1 or Category 2 Computational Load Owner/Operator registration categories.**
- **“Category 1 CLO/CLOP”** should be aligned with the BES criteria, which would be entities who own/operate Computational Load Facilities (CLFs) that are **greater than or equal to 75 MW** of Computational Load connected at **greater than or equal to 100 kV.**
- **“Category 2 CLO/CLOP”** should be aligned with the lower threshold value supported by NERC and industry, which would be entities who own/operate Computational Load Facilities that are **greater than or equal to 20 MW** of Computational Load connected at **greater than or equal to 60 kV.**

This structure would allow the Electric Reliability Organization enterprise to focus its initial entity registration, Standards Development efforts, and compliance education activities on the highest-risk CLOs/CLOPs first. This framework also establishes a more manageable path to subsequently address lower-risk Category 2 CLOs/CLOPs in future work phases.

NERC Rules of Procedure Team —

AIONTRACKS, Inc. respectfully submits the following comments on the proposed Rules of Procedure changes establishing the Computational Load Entity (CLE) functional entity type and the registration criteria in proposed Appendix 5B.

Comment 1 — The proposed registration criteria are well-calibrated for the threshold class.

The proposed criteria — aggregate connected load capability ≥ 20 MW at a single point of interconnection at ≥ 60 kV, hosting ≥ 1 MW of computational load—appear well-aligned with the BPS-impact threshold for which mandatory Reliability Standards are appropriate. AIONTRACKS supports the proposed thresholds.

Comment 2 — The criteria may benefit from a class-of-service descriptor captured at registration.

Computational loads vary substantially in workload composition. A facility hosting predominantly artificial intelligence training workloads has materially different electrical and operational characteristics from facility hosting predominantly inference workloads, or from a mixed-workload facility. The class of service hosted at the facility affects both the demand profile under normal conditions and the response capability during disturbances. NERC may wish to consider whether the registration form should capture a class-of-service descriptor (e.g., training-dominant, inference-dominant, mixed, batch-dominant, or other) to support downstream standard application and audit. Capturing this information at registration imposes minimal burden on the registrant and provides NERC and the Regional Entities with material context for compliance review.

Comment 3 — The criteria should anticipate facilities hosting computational loads on behalf of multiple end-users.

Some computational load facilities operate as multi-tenant infrastructure, hosting computational workloads on behalf of multiple end-user customers under contractual arrangements. The proposed criterion that the CLE must be "the end-user or the entity that hosts end-users that receives electric power" appropriately captures this case, but NERC may wish to clarify in guidance whether the multi-tenant facility operator is the registered CLE or whether each end-user is registered separately. Clarification of this point will reduce registration ambiguity for the operator class likely to host the largest aggregate computational load.

AIONTRACKS appreciates the opportunity to comment and offers to serve as a technical resource to NERC staff in finalizing the registration criteria.

Respectfully,

Praneet Sharma, M.D.
Founder and Chief Executive Officer
AIONTRACKS, Inc.
Phoenix, Arizona

Date: April 30, 2026

North American Electric Reliability Corporation
3353 Peachtree Road NE, Suite 600N
Atlanta, Georgia 30326

Re: Comments of Alphataraxia on Proposed Revisions to NERC Rules of Procedure Appendices 2, 5A, and 5B
(Computational Load Entity Registry Criteria)

Dear NERC Staff:

Alphataraxia Management LP (“**ATX**”) submits these comments on NERC’s April 1, 2026 posting of proposed revisions establishing a “Computational Load Entity” (“**CLE**”) functional entity type. ATX is engaged in energy trading in domestic ISO/RTO markets and, as a wholesale market participant, has a direct interest in the accuracy of load data and planning models that underpin locational marginal prices, congestion revenue rights, and reserve margin determinations. Accurate classification and registration of computational loads is not merely a technical compliance matter—as large-scale data center and AI infrastructure loads increasingly stress transmission systems and distort reserve margin calculations, failures to properly account for their magnitude and consumption profiles risk cascading reliability events that ultimately harm end-use consumers through price volatility, inadequate capacity procurement, and, in the most severe cases, supply shortfalls. ATX supports the CLE initiative and submits these comments to urge NERC to require, as a registration condition, transparency about load qualification criteria and disclosure of historical consumption data.

I. Background

In September 2025, NERC issued its Level 2 Industry Recommendation on Large Load Interconnection, Study, Commissioning, and Operations. The Level 2 Alert responded to a series of grid disturbances in which more than 1,000 MW of large-load output was unexpectedly reduced across multiple events in 2024 and 2025. NERC’s Aggregated Report on the Level 2 Alert responses (the “**Aggregated Report**”) identified a central, recurring deficiency: load data for large commercial and industrial customers is “often incomplete, aggregated, or lacks necessary granularity, such as hourly or daily usage trends for use in short-term load forecasting.” The LLWG’s March 2026 white paper confirmed that existing Reliability Standards and industry processes are inadequate for reliable large load integration. The proposed registry criteria represent the correct structural location to address this deficiency—before downstream standards development locks out a data-first approach.

II. Historical Consumption Data Must Be a Registration Condition

NERC has recognized that computational loads can exhibit electrical behavior that is “distinctive and often less predictable than conventional loads.” The Aggregated Report confirmed that forward-looking projections submitted at interconnection bore little resemblance to actual operational profiles. AI training and high-

performance compute workloads have been specifically identified as among the load types most likely to introduce forced oscillations into the grid.

ATX recommends that the criteria require CLEs, as a condition of initial registration, to commit to submitting metered consumption data at hourly intervals, including observed peak demand, minimum demand, and average load factor. For newly commissioned facilities, the most complete available observed data plus a validated dynamic load model should be required, as well as a commitment to post metered consumption data to a nationwide database that is publicly available. Public posting of such data has huge value. There is already a nationwide database for generators posted by the US Energy Information Administration (available at <https://www.eia.gov/electricity/data.php>), which has supported significant analysis and research; a similar database for CLEs would add immense value to research into reliability and optimization.

III. Load Modeling and Market Integrity Require Actual Consumption Data

Level 2 Alert Recommendation 5 directed Transmission Planners, Resource Planners, and Planning Coordinators to incorporate large loads into near-term and long-term demand forecasts, accounting for intra-minute variability, cyclical ramping, and aggregate load disconnections during contingencies. That recommendation presupposes access to actual consumption data. Without it, the planning models that underpin wholesale market prices in PJM, MISO, SPP, and ERCOT will continue to degrade in accuracy as CLE penetration grows.

When load models do not reflect actual CLE consumption profiles, the resulting distortions affect locational marginal prices, create unanticipated transmission congestion, and affect the financial settlement of FTRs and virtual transactions. ATX accordingly recommends that the criteria explicitly require CLEs to provide historical consumption data to a nationwide database.

IV. Transparent Qualification Criteria Are Necessary to Prevent Registration Arbitrage

Opaque CLE qualification criteria create incentives for registration arbitrage. Entities with borderline facilities, like data centers co-located with conventional operations, multi-metering-point campuses, or facilities with significant on-site backup generation, have incentives to structure their operations to minimize reported interconnection load and avoid registration, reducing compliance costs while continuing to affect grid conditions. NERC should publish detailed guidance on how the 20 MW aggregate connected load threshold is measured across shared substations, co-located generation, and mixed-use facilities, and how “single point of interconnection” is applied to campus configurations. Consistent, data-grounded application across all Regional Entities is essential to preventing regulatory gaps and preserving the competitive integrity of wholesale markets.

As data centers are projected to be more than 10% of total U.S. electricity consumption by 2028, the aggregate market impact of unmodeled or under-modeled CLE consumption will be substantial. Wholesale market participants that hold financial transmission rights and manage load-serving obligations in CLE-concentrated load zones are entitled to rely on planning models that reflect actual CLE consumption behavior.

V. Conclusion

ATX urges NERC to strengthen the proposed CLE registry criteria by incorporating the idea of a nationwide database of CLEs. In particular, (i) requiring submission of consumption data at hourly intervals; (ii) defining load qualification criteria with sufficient specificity to enable consistent Regional Entity application and prevent registration arbitrage; and (iii) explicitly requiring that historical consumption data flow be provided so that it can be posted publicly (ideally, in a national database) so as to allow Transmission Planners, Planning Coordinators, Balancing Authorities, and RTOs/ISOs to incorporate into planning models. These measures are necessary to give practical effect to the Aggregated Report's findings, to enable existing registered entities to fulfill their planning standards obligations, and to preserve the competitive integrity of the wholesale markets NERC's reliability framework supports.

Respectfully submitted,

Brian Kozlowski
General Counsel
Alphataraxia Management LP

Ameren comments on the NERC Proposed Computational Load Entity ROP Revisions

Ameren agrees with EEI's comments on the revisions.

May 15, 2026

North American Electric Reliability Corporation
Rules of Procedure Comments

Via Electronic Submission: ROPcomments@nerc.net

**Re: Comments on Proposed Revisions to NERC Rules of Procedure — Computational Load Entity
(Appendices 2, 5A, and 5B)**

I. Introduction

Arevon Energy, Inc. (Arevon) appreciates the opportunity to submit comments on the proposed revisions to the North American Electric Reliability Corporation (“NERC”) Rules of Procedure (“RoP”) establishing a new Computational Load Entity (“CLE”) functional category and associated registration criteria in Appendices 2, 5A, and 5B. These comments are submitted in response to NERC’s April 1, 2026 posting and the 45-day public comment period closing May 15, 2026.

Arevon recognizes that emerging large computational loads — including data centers, AI compute clusters, and cryptocurrency mining facilities — present genuine reliability challenges that require proactive action. While NERC’s Large Loads Action Plan objectives are valid, we urge NERC to refine the proposed RoP revisions to ensure that (1) definitions are precise and risk-calibrated, (2) registration thresholds capture only those entities whose behavior drives the identified reliability risks, (3) the obligations that follow registration are proportionate and appropriately assigned, and (4) the implementation timeline is realistic for entities that are entirely new to NERC’s compliance framework.

The comments included in this document are organized into four sections: Appendix 2 (Definitions), Appendix 5A (Registration Process), Appendix 5B (Registry Criteria), and Cross-Cutting Process and Implementation Considerations.

II. Appendix 2 — Definitions Used in the Rules of Procedure

A. The Proposed “Computational Load” Definition Requires Greater Precision

As proposed, “Computational Load means Load comprised of electric power demand from information technology equipment, such as servers, storage, and networking hardware.” While we agree with NERC staff that a term other than “Large Load” is appropriate — since “Large Load” can connote many different types of loads on the Bulk Power System (“BPS”), such as large process facilities, education and medical campuses, pharmaceutical loads, and manufacturing loads — the proposed definition introduces several ambiguities that should be resolved before adoption.

1. The phrase “such as” is non-inclusive and open-ended. It creates uncertainty about what equipment falls within scope. Virtually all BPS-connected facilities may contain some information technology equipment (PLCs, RTUs, SCADA systems, routers, personal computers). Without a clear, inclusive list or a quantitative threshold tied to the proportion of IT-related demand, the definition risks capturing facilities that were never the intended subject of NERC’s Large Load Action Plan. NERC should replace “such as” with “including, but not limited to” or provide a closed-form definition anchored to the types of facilities actually driving reliability risk (data centers, AI compute clusters, cryptocurrency mining).

2. “Storage” should be amended to “data storage.” As drafted, “storage” is ambiguous and could be interpreted to include Battery Energy Storage Systems (“BESS”), which are separately governed under existing NERC registration criteria for inverter-based resources. The intent is clearly to reference Network Attached Storage (“NAS”), Hard Disk Drives (“HDDs”), and Solid-State Drives (“SSDs”). This clarification is essential to avoid unintended overlap with generation-side registration criteria.

3. The term “Computational” may be too narrow. Some referenced equipment — particularly networking hardware — has minimal computational capability compared to CPUs and GPUs. As power electronic equipment (e.g., solid-state transformers) is deployed at customer sites, the definition may become even more misaligned with the actual risk drivers. NERC should consider a broader term such as “Electronic Load,” consistent with ERCOT’s adoption of “Large Electronic Load” in NPRR 1308, which extends beyond purely computational activities and better captures the full spectrum of loads driving the identified reliability risks.

B. The Definition Excludes Non-IT Large Loads with Identical Reliability Risks

The proposed definition captures only loads from information technology equipment. Yet NERC’s own Large Loads Working Group (“LLWG”) reports identify hydrogen electrolyzers, industrial heat pumps, advanced manufacturing facilities, and other energy-intensive loads as posing identical reliability risks — rapid demand changes, voltage sensitivity, and forecasting difficulty.

By limiting the definition to “information technology equipment,” NERC would need to repeat the entire RoP revision process when these load types reach scale — delaying the reliability protections that the Large Loads Action Plan was designed to deliver. NERC should either broaden the definition now to encompass all large loads exhibiting the identified risk behaviors, or explain in its technical reasoning why the identified reliability risks are unique to computational equipment and do not apply to other emerging large loads.

C. The “Computational Load Entity” Definition Has Structural Ambiguities

As proposed, “Computational Load Entity means the end-user or the entity that hosts end-users that receives electric power for Computational Load.” This definition raises several structural concerns:

1. Owner/Operator Distinction. NERC’s existing registration framework distinguishes between owners and operators (e.g., GO/GOP, TO/TOP). Many computational load facilities will have separate ownership and operating structures — particularly in the colocation and multi-tenant data center market. The proposed single “Computational Load Entity” functional type does not accommodate this reality and should adopt a parallel Owner/Operator structure (e.g., CLO/CLOP) to properly assign differentiated reliability responsibilities.

2. Ambiguity in “end-user or the entity that hosts end-users.” Where both an end-user and a hosting entity exist at a single facility, the definition does not clarify which party bears registration obligations — or whether both must register. This is a critical gap given the prevalence of colocation and multi-tenant data center arrangements.

3. Potential Third-Party Registration Obligations. The use of “end-user” could introduce lessee or tenant entities as separately registrable parties, creating a three-tier registration structure (owner, operator, tenant) that is unprecedented in NERC’s framework and operationally unworkable without further guidance. NERC should clarify whether registration follows facility ownership, operational control, or electrical metering boundaries. For example, if Company A and Company B lease a warehouse owned by Company C that hosts the equipment creating the demand for electricity (load), who should register? Company A, B, or C or all of them? There could be several permutations and combinations of such scenarios.

D. The “Users of the BPS” Legal Authority Does Not Compel Registration of All Load Customers

NERC’s Technical Reasoning grounds the proposed CLE registration in the Federal Power Act’s grant of authority over “users, owners, and operators of the Bulk Power System.” 16 U.S.C. § 824o, Section 215(b); 18 C.F.R. § 39.2(c) (2026). However, the existence of jurisdictional authority does not, by itself, determine where the registration threshold should be set. NERC acknowledges this materiality principle by proposing thresholds rather than blanket registration of all BPS-connected loads. The critical question, therefore, is not whether NERC has authority to register computational load entities — it does — but whether the proposed thresholds are appropriately calibrated to capture entities whose behavior materially impacts BPS reliability. As discussed in Section IV below, the proposed 1 MW Computational Load threshold is set so low that it captures entities whose electrical behavior does not implicate the reliability risks identified in NERC’s own technical analysis.

III. Appendix 5A — Organization Registration and Certification Manual

A. The Registration Process Requires a Reasonable Transition and Implementation Timeline

The addition of “Computational Load Entity (CLE)” as a registrable entity type represents a historic first. The registration process should include a reasonable implementation period and clear guidance on the registration workflow, particularly for entities that have no institutional familiarity with NERC’s compliance obligations, audit expectations, or the ERO Enterprise framework. It is recommended that a clear communication be developed for outreach to entities that would potentially fall under this registration type, as well as the ecosystem supporting those entities.

B. Registration Must Be Coordinated with Project 2026-02 Standards Development

Registration should not outpace the development of applicable Reliability Standards. Registering CLEs before the standards they must comply with are finalized creates a compliance gap — entities would be on the NERC Compliance Registry with no clear obligations, or worse, subject to enforcement actions for standards still under development. NERC’s Standards Committee accepted the Project 2026-02 SAR on March 18, 2026, and a Standards Drafting Team has been appointed. The current project timeline targets Board adoption in December 2026. Accordingly, the effective date for CLE registration should be no earlier than the effective date of the first mandatory Reliability Standard applicable to CLEs.

C. A Clear De-Registration Process Is Needed

The proposed criteria define when an entity must register but are silent on what happens when an entity falls below the threshold — e.g., a facility reduces IT load below 1 MW, or aggregate load drops below 20 MW due to decommissioning or facility reconfiguration.

Appendix 5A should include a clear de-registration process for CLEs that no longer meet the threshold criteria. The current RoP provisions for de-registration of other functional entities (e.g., GOs, TOPs) may not translate directly to CLEs, given the dynamic nature of computational load facilities. NERC should specify the evidentiary requirements, timeline, and Regional Entity review process for CLE de-registration to avoid indefinite registration of entities that no longer pose the identified reliability risks.

IV. Appendix 5B — Statement of Compliance Registry Criteria

A. Registration Thresholds Are Misaligned with Actual Reliability Risk

As proposed, the three-part threshold requires: (1) aggregate connected Load capability $\geq$ 20 MW; (2) at a single point of interconnection to the BPS at $\geq$ 60 kV; and (3) hosting $\geq$ 1 MW of Computational Load. This threshold captures facilities that do not present the reliability risks identified in NERC's own technical analysis. It also appears that these numbers have simply been adopted from the IBR registration initiative (aggregation and interconnection thresholds), rather than based on any solid engineering or technical analysis demonstrating risk to the BPS.

The 1 MW Computational Load threshold is disproportionately low. At 20 MW aggregate load, 1 MW of Computational Load represents just 5% of total facility demand. A 75 MW industrial facility with 1 MW of IT equipment for servers, PLCs, RTUs, routers, and personal computers would be required to register despite presenting none of the rapid load-swing, voltage stability, or protection coordination risks documented in the LLWG white papers and the Level 3 Alert. Additionally, current Registered Entities that may have 1 MW of computational load may also fall under this category – does that mean they would have to register as a CLE as well and have the additional administrative burden and cost of complying with standards that may be applicable to a CLE?

ERCOT's approach may better reflects risk. ERCOT's proposed NPRR 1308 definition requires that 50% or greater of site demand consist of power-electronic-based computational load, with a minimum aggregate peak demand of 75 MW. NERC's criteria, intended to establish the minimum acceptable continent-wide requirements, may conflict with ISO/RTO- has determined appropriate for its footprint.

Recommendation. NERC should consider surveying different markets or transmission organizations that are interconnecting such large loads to determine what the right threshold (e.g., Computational Load percentage of 25–50% of aggregate demand) for registration is to ensure only facilities whose electrical behavior genuinely poses BPS reliability risks — namely, those with high concentrations of power-electronic-based load capable of rapid, large-magnitude demand changes are registered. This approach would align with the LLWG's findings and permit transmission organizations to adopt more stringent criteria as appropriate for their systems.

The aggregate impact doctrine requires careful calibration for load entities. NERC's Technical Reasoning invokes the aggregate impact doctrine, citing 150 FERC ¶ 61,213 at P 58 (2015), which provides that "Registry Criteria provide that a class of entities, each of which would be individually excluded, may nevertheless be registered based on their aggregate impact on Bulk-Power System reliability." However, the aggregate impact doctrine was originally developed in the context of generation resources and was subsequently applied to inverter-based resources (IBRs) under the Category 2 GO/GOP registration initiative. Applying this doctrine to load entities requires careful calibration. The aggregate impact of a class of computational load entities may not justify registering individual entities within that class whose

own characteristics do not pose the identified reliability risks. A 20 MW mixed-use industrial facility with 1 MW of IT/computational load equipment may not exhibit the power electronic load (“PEL”) behavior that NERC identifies as the risk driver — regardless of how many such facilities exist in aggregate.

The 1 MW threshold does not effectively serve as a de minimis filter. NERC’s Technical Reasoning states that “the intent behind the ‘1 MW or greater’ of Computational Load threshold is to distinguish that those entities, such as industrial manufacturers, with a ‘de minimus’ amount of Computational Load do not meet the criteria for registration as a Computational Load Entity.” However, 1 MW does not effectively serve this de minimis filtering function. Modern industrial facilities routinely may contain well over 1 MW of information technology equipment — including server rooms, SCADA systems, industrial automation platforms, edge computing infrastructure, and enterprise networking. A large petrochemical complex, steel mill, or semiconductor fabrication facility with 1–5 MW of IT equipment would be captured by the proposed criteria despite presenting none of the rapid load-swing or voltage sensitivity risks associated with data centers. A percentage-based threshold (e.g., 25–50% of aggregate demand consisting of Computational Load) would more effectively distinguish facilities whose primary function is computational from facilities that simply use computing equipment ancillary to their primary industrial process.

The registration criteria should align with NERC’s own technical risk analysis. Notably, NERC’s Technical Reasoning identifies power electronic load (“PEL”) behavior as the actual technical risk driver, citing the LLWG’s finding that “many new large loads are [power electronic loads] (PELs) that use software controls to manage the load behavior rather than the mechanical controls used by older motor loads” and that “PEL control systems enable loads to rapidly change electricity consumption, creating previously unseen load behavior that can cause BPS planning and operational challenges.” Yet the proposed definition and registration thresholds are equipment-type-based, not behavior-based. NERC should align its registration criteria with its own technical reasoning by incorporating behavioral characteristics — such as the proportion of PEL-based load, rate-of-change capability, or voltage sensitivity — rather than simply counting megawatts of “information technology equipment.” This approach would more precisely target the entities whose behavior drives the identified reliability risks.

B. A Tiered Registration Approach Should Reflect Non-Linear Risk Scaling

The LLWG’s own analysis concluded that “the potential impact of several 100 MW loads may be less than the impact of a 500 MW load and may require different treatment” and that “the emergence of gigawatt-scale loads may create exponential complications for study, planning and operations.” Yet the proposed criteria apply a single binary threshold — register or don’t.

The registry criteria should support a tiered or graduated approach to compliance obligations, consistent with the LLWG’s finding that risk scales non-linearly with load size. A 25 MW data center and a 500 MW hyperscale campus present fundamentally different risk profiles, yet the proposed criteria would subject both to the same registration status. Consider distinguishing between load sizes (e.g., 20–100 MW, 100–300 MW, >300 MW) to enable proportional obligations in the standards that follow.

C. NERC Should Explicitly Preserve Regional Flexibility — Floor vs. Ceiling

Multiple ISOs/RTOs are developing their own large load frameworks with materially different thresholds. The following table illustrates the divergence:

ISO/RTO	Threshold	Key Feature	Activity Level
ERCOT (NPRR 1308)	≥ 75 MW; ≥ 50% power-electronic load	Higher bar, behavior-based	High
SPP (HILL)	≥ 10 MW at ≤ 69 kV; ≥ 50 MW at > 69 kV	Voltage-tiered	High
PJM	Co-location order; CIFP underway	Service-type based	High
NERC (Proposed)	≥ 20 MW aggregate; ≥ 1 MW CL at ≥ 60 kV	Lowest threshold	—

NERC’s proposed criteria establish a continent-wide floor, but they should explicitly state that Regional Entities and ISOs/RTOs retain the ability to adopt more stringent thresholds and additional criteria appropriate to their system conditions — consistent with NERC’s traditional approach to registration. Without this clarification, the proposed criteria could be misinterpreted as preempting more tailored regional approaches like ERCOT’s NPRR 1308 or SPP’s HILL framework.

D. “Single Point of Interconnection” Requires Clarification for Co-Located Facilities

The criterion requiring connection “at a single point of interconnection to the Bulk Power System at a voltage greater than or equal to 60 kV” does not address facilities with multiple POIs, shared substations, or co-located generation. NERC should clarify:

- Whether aggregation occurs across multiple POIs at the same physical site;
- How co-located generation (solar, BESS) and computational load at a shared POI are treated for purposes of the 20 MW aggregate calculation;
- Whether behind-the-meter generation offsets are considered — i.e., whether a 30 MW data center with 15 MW of on-site solar has an “aggregate connected Load capability” of 30 MW or 15 MW; and
- How phased construction or modular expansion of computational load facilities affects the timing of registration obligations.

E. Registration Thresholds Should Align with the Level 3 Essential Action Alert

On May 4, 2026, NERC issued a Level 3 Essential Action Alert identifying seven essential actions that registered entities must take to address immediate risks posed by computational loads interfacing with the BPS:

1. Develop modeling requirements for computational loads;
2. Complete system studies and planning improvements;
3. Update definition of “qualified change”;
4. Establish a commissioning process for computational loads;
5. Implement protection and coordination enhancements;
6. Install dynamic fault recording devices; and
7. Create Interpersonal Communication capabilities.

These seven essential actions provide the clearest preview yet of the obligations CLEs will face. NERC should explain how the proposed registration thresholds were calibrated to capture the entities whose behavior drives the need for these essential actions. If the thresholds capture entities whose facilities do not implicate these actions (e.g., a 20 MW mixed-use industrial facility with 1 MW of IT equipment), the criteria are overbroad and should be revised.

Notably, the two specific incidents that underpin NERC’s urgency — the January 2025 Eastern Interconnection large load loss event and the January 2026 Texas Interconnection voltage-sensitive crypto load reductions — both involved facilities with high concentrations of computational load, not mixed-use facilities with incidental IT equipment. This further supports a higher percentage threshold that targets the facilities whose electrical behavior actually caused these events.

V. Cross-Cutting Process and Implementation Considerations

A. Protect Against Unintended Scope Creep to Generator Owners and Operators

Any new registration criteria and the reliability standards that follow must clearly delineate which risks are attributable to computational load behavior versus generation performance. Risks driven by computational load characteristics — volatility, forecasting uncertainty, rapid load changes, and protection coordination challenges — should be addressed through CLE-specific obligations, not through expanded requirements on Generator Owners or Generator Operators who do not control load behavior.

Furthermore, any new or revised Reliability Standards developed as a result of these RoP revisions should be applied prospectively and aligned with approved interconnection agreements, study

assumptions, and existing facility designs. Retroactive application would introduce compliance risk without corresponding reliability benefit, particularly for co-located generation and storage facilities.

B. Jurisdictional Coordination with FERC's ANOPR (Docket RM26-4-000)

NERC's proposed CLE registration criteria are advancing in parallel with FERC's Advanced Notice of Proposed Rulemaking ("ANOPR") under Docket RM26-4-000, initiated pursuant to DOE Secretary Wright's rare invocation of Section 403 authority. The ANOPR will establish its own federal large load interconnection framework, including standardized study procedures, cost allocation, and protection system requirements for loads greater than 20 MW connecting to transmission.

NERC should explain how CLE registration will coordinate with — and avoid conflicting with — FERC's emerging interconnection requirements. The ANOPR's > 20 MW threshold matches NERC's proposed aggregate load threshold, but FERC's scope extends to all large loads, not just computational loads. Entities may face overlapping, inconsistent, or duplicative obligations unless NERC and FERC explicitly coordinate these parallel proceedings. At minimum, NERC's filing with FERC for approval of the RoP revisions should address how the two frameworks will interact. Notably, NERC itself filed comments in Docket RM26-4-000 on November 21, 2025, acknowledging the need for coordination between NERC's reliability framework and FERC's interconnection requirements for large loads. The proposed RoP revisions should be consistent with the positions NERC articulated in those comments.

C. Data Confidentiality Protections for CLE Operational Data

Many CLEs — particularly hyperscale data centers — consider their power consumption, load profiles, demand forecasts, and facility configurations to be commercially sensitive and competitively valuable. The proposed RoP revisions do not address how CLE operational data will be protected under NERC's existing confidentiality framework.

The revisions should clarify how data submitted by CLEs will be treated under NERC's confidentiality classifications (Public, Limited Disclosure, Confidential, Restricted-Confidential). Without clear data protection commitments, entities may resist registration or provide incomplete data, undermining the reliability objectives of the Large Loads Action Plan. NERC should develop CLE-specific data handling protocols in coordination with the Regional Entities before the registration process begins.

D. NERC Should Publish a Clear Implementation Roadmap

The Level 3 Essential Action Alert (May 4, 2026), the Voluntary Reliability Guideline, the RoP registration criteria (comment period closing May 15, 2026), and the Phase 1 bridge standard from Project 2026-02 are all advancing in parallel on aggressive timelines. NERC should publish a clear implementation roadmap that sequences registration, standard development, and enforcement expectations to avoid confusion and premature compliance exposure.

Additionally, NERC and the Regional Entities should develop targeted onboarding materials, compliance workshops, and a reasonable ramp-up period before enforcement actions are pursued against newly

registered CLEs. These entities are, by definition, new to NERC’s compliance framework and will require structured guidance to achieve meaningful compliance.

VI. Conclusion

The Large Loads Action Plan represents an important and timely initiative. However, the proposed RoP revisions require refinement in several critical areas to ensure that the new CLE registration framework achieves its intended reliability objectives without creating unintended consequences.

In summary, the comments are organized around four key themes:

1. **Definitional Precision:** The “Computational Load” and “Computational Load Entity” definitions require greater specificity to avoid overbroad application, resolve ambiguities regarding ownership/operating structures, and close gaps for non-IT large loads with identical reliability risk profiles.
2. **Risk-Calibrated Thresholds:** The Appendix 5B registration criteria should be revised to increase the Computational Load percentage threshold, adopt a tiered approach reflecting the LLWG’s finding that risk scales non-linearly with load size, incorporate behavioral characteristics consistent with NERC’s own PEL-based technical reasoning, and explicitly preserve regional flexibility for ISOs/RTOs to adopt more stringent criteria.
3. **Implementation Sequencing:** CLE registration should not outpace standards development or enforcement readiness. NERC should coordinate registration timing with Project 2026-02, FERC’s ANOPR under Docket RM26-4-000, and the Level 3 Alert essential actions to avoid overlapping or conflicting obligations.
4. **Scope Protection:** Risks attributable to computational load behavior should be addressed through CLE-specific obligations — not through expanded requirements on Generator Owners or Generator Operators who do not control load behavior.



Notice of Proposed Revisions to the NERC Rules of Procedure and Request for Comments – Computational Load Entity

NERC Rules of Procedure Appendix 2 (Definitions Used in the Rules of Procedure); Appendix 5A (Organization Registration and Certification Manual); and Appendix 5B (Statement of Compliance Registry Criteria)

[Proposed Changes to Rules of Procedure](#)

[Appendix 2 Clean](#) | [Redline](#)

[Appendix 5A Clean](#) | [Redline](#)

[Appendix 5B Clean](#) | [Redline](#)

[Summary of Changes](#)

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ROPcomments@nerc.net.

The proposed revisions address recommendations from the NERC Large Loads Working Group (a subgroup of the NERC Reliability and Security Technical Committee) whitepaper “[Assessment of Gaps in Existing Practices, Requirements, and Reliability Standards for Emerging Large Loads](#)” and are further supported by the findings in the “[NERC Aggregated Report on NERC Level 2 Industry Recommendation: Large Load Interconnection, Study, Commissioning, and Operations](#)” analyzing responses to a [Level 2 Alert](#) on the interconnection of large loads. The recommendations and findings indicate a need for a new registered entity to comply with Reliability Standards (currently under development in [Project 2026-02 – Computational Loads](#)) to reduce risks to the reliability of the Bulk Power System.

In pursuing its mission to assure the effective and efficient reduction of risks to the reliability and security of the grid, NERC works with owners, operators, users, and consumers as well as federal, state, and provincial regulators across North America. To that end, NERC welcomes input from all stakeholders on the proposed revisions and values the opportunity to engage with stakeholders on how the proposed revisions can contribute to the reliability of the Bulk Power System.

For more information or assistance, please contact [Marisa Hecht](#) via email or at **404-446-9620**.

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Comments on Proposed Revisions to NERC ROP: Computational Loads

ATC generally supports the comments presented by the Trade Associations (APPA, EEI, LPPC, NRECA, TAPS, and NARUC) including the Section 1 ‘Overarching Principles’; maintaining consistency with other NERC Registration Categories; defining Computational Load to avoid inadvertent inclusion of previously stable large loads like industrial and commercial processes; and Section 4 ‘Comments on NERC’s Proposed Revisions’.

Additional comments are provided below.

ATC Proposed Modifications

1. Add a new definition Computational Load Facility (CLF):

“Computational Load Facility” means a collection of electrical equipment located at one or more buildings, structures, or installations that is designed to receive power, has an aggregate connected maximum peak demand greater than or equal to 20 MVA, hosts 50% or greater of the peak demand load as Computational Load, and is served through electric facilities at a voltage level greater than or equal to 60 kV from one or more points of connection.

A clear definition of a Computational Load Facility is essential for Project 2026-02 success. Existing standards that currently address the same types of essential actions being proposed for data centers but are applicable to other comparable equipment (like generators) consistently rely on a Facility-type definition. This is consistent with how Generator Owner/Operator and Generating Facility are treated, defined, and included in the registration criteria.

Data center sites can have a significant number of physical and electrical configurations. For example, many data center sites can consist of one or more buildings, with each building having one or more connections to the Bulk Power System at the same location. In some cases, data center owners may consider that each building is an individual data center, while in other cases the collection of buildings connected to the grid is viewed as the data center. The risk to the BES comes from the aggregate load behavior of facilities at a single site. This was emphasized in NERC’s “Characteristics and Risks of Emerging Large Loads” whitepaper that, while specific load types and thresholds to define Large Loads could not be defined across industry at that time, all agreed that any such definition should include “aggregation of load facilities at a single site behind one or more point(s) of interconnection that can pose reliability risks to the BPS.”

2. Replace CL Entity (CLE) with CL Operator (CLOP) and CL Owner (CLO)

“Computational Load Operator”: means the end-user entity that operates a Computational Load Facility, and/or the entity that operates a Computational Load Facility that hosts end-users that receives electric power at a Computational Load Facility.

Comments on
Proposed Revisions to NERC ROP: Computational Loads

“Computational Load Owner”: means the end-user entity that owns and maintains a Computational Load Facility, **and/or** the entity that owns and maintains a Computational Load Facility that hosts end-users that receives electric power **at a Computational Load Facility**.

The risk to the BPS and the gaps in NERC reliability standards, as noted in NERC’s “Assessment of Gaps in Existing Practices, Requirements, and Reliability Standards for Emerging Large Loads” is driven by Computational Load impacts and behavior, not other type of load, and Computational Load Entity should focus only on the amount of data-centric/information technology type of load.

3. Clarify the Computational Load (CL) definition:

“Computational Load” means Load comprised **primarily** of electric power demand from **power electronic-based** information technology equipment.

4. Modify Computational Load criteria thresholds in Appendix 5B

ATC is concerned that NERC’s Appendix 5B definition and criteria for Computational Load Entity is too broad and could unintendedly include larger loads that do not pose the same reliability risks as data centers. These existing non-data center facilities are not the target of these to address the most critical reliability issues associated with integrating these large loads onto the BPS in the near-term. It is inefficient to require all facilities 20 MW and greater with 1 MW of computation load to register, and difficult to then depend on the standard to further delineate between hospitals and data centers for requirements in new or existing standards.

Therefore, ATC proposes that the criteria and definition for Computational Load Entity be modified to “**1) contributes to an aggregate connected maximum peak demand greater than or equal to 20 MVA; 2) hosts 50% or greater of the peak demand load as Computational Load; and 3) is served through electric facilities at a voltage level greater than or equal to 60 kV from one or more points of connection**”

ATC also recommends that NERC considers requiring each applicable Transmission Operator and Transmission Owner to perform an entity specific study to determine the MW threshold at which a computational load presents reliability risk within its system.

ATC would consider supporting the Category 1/Category 2 designations proposed by the Trade Associations as long as both categories are subject to the same requirements and both categories are implemented without delay between them.

AZPS Comments on Proposed Revisions to NERC ROP: Computational Load Entity

May 8, 2026



Notice of Proposed Revisions to the NERC Rules of Procedure and Request for Comments – Computational Load Entity

NERC Rules of Procedure Appendix 2 (Definitions Used in the Rules of Procedure); Appendix 5A (Organization Registration and Certification Manual); and Appendix 5B (Statement of Compliance Registry Criteria)

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The proposed revisions address recommendations from the NERC Large Loads Working Group (a subgroup of the NERC Reliability and Security Technical Committee) whitepaper “[Assessment of Gaps in Existing Practices, Requirements, and Reliability Standards for Emerging Large Loads](#)” and are further supported by the findings in the “[NERC Aggregated Report on NERC Level 2 Industry Recommendation: Large Load Interconnection, Study, Commissioning, and Operations](#)” analyzing responses to a [Level 2 Alert](#) on the interconnection of large loads. The recommendations and findings indicate a need for a new registered entity to comply with Reliability Standards (currently under development in [Project 2026-02 – Computational Loads](#)) to reduce risks to the reliability of the Bulk Power System.

In pursuing its mission to assure the effective and efficient reduction of risks to the reliability and security of the grid, NERC works with owners, operators, users, and consumers as well as federal, state, and provincial regulators across North America. To that end, NERC welcomes input from all stakeholders on the proposed revisions and values the opportunity to engage with stakeholders on how the proposed revisions can contribute to the reliability of the Bulk Power System.

For more information or assistance, please contact [Marisa Hecht](#) via email or at **404-446-9620**.

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AZPS Comments on Proposed Revisions to NERC ROP: Computational Load Entity

May 8, 2026

AZPS appreciates the opportunity to provide comments on the draft proposed revisions to the NERC Rules of Procedure: Computational Loads. AZPS respectfully requests additional clarification to ensure the computational load definition and registration criteria appropriately include all applicable entities, including owners, operators of large loads, and host-end users.

AZPS supports EEI's comments with the following exceptions and proposed modifications:

- AZPS agrees with EEI that a Computational Load Facility definition is needed and supports EEI's proposal with the caveat that AZPS believes the voltage should be 60kV.
 - **"Computational Load Facility"** means a collection of electrical equipment located at one or more buildings, structures, or installations that is designed to receive power, has an aggregate connected maximum peak demand of Computational Load greater than or equal to 75 MW, and is served through electric facilities at a voltage level greater than or equal to **60 kV** from one or more points of connection.
- AZPS agrees with EEI that a Computational Load Owner and Computational Load Operator definition is needed to clearly define ownership and responsibilities.
- AZPS is of the position that clarity is needed in the registry criteria language to ensure that Large Load entities that are embedded in small distribution entities such as a cooperative, municipal or other retail service utilities that serve Computational Load Entities should be applicable to registration criteria and not be excluded from Reliability requirements. All entities (including small distribution entities such as cooperatives, municipalities or other retail service utilities) with computational loads should be required to meet the established thresholds.
- AZPS is of the opinion that the proposed Statement of Compliance Registry Criteria for Computational Load Entity for hosts be increased from 1MW to 20MW or greater of Computational. AZPS submits the following proposed modification: A Computational Load Entity that 1) contributes to an aggregate connected Load capability greater than or equal to 20 MW; 2) at a single point of interconnection to the Bulk Power System at a voltage greater than or equal to 60 kV; and 3) hosts 20MW or greater of Computational Load.

AZPS supports the following comments submitted by EEI on behalf of their members:

The Trades propose the following modifications:

1. **Add a new definition Computational Load Facility (CLF):**

AZPS Comments on Proposed Revisions to NERC ROP: Computational Load Entity

May 8, 2026

“Computational Load Facility” means a collection of electrical equipment located at one or more buildings, structures, or installations that is designed to receive power, has an aggregate connected maximum peak demand of Computational Load greater than or equal to 75 MW, and is served through electric facilities at a voltage level greater than or equal to 100 kV from one or more points of connection.

A clear definition of a Computational Load *Facility* is essential for Project 2026-02 success. Existing standards that currently address the same types of essential actions being proposed for data centers but are applicable to other comparable equipment (like generators) consistently rely on a *Facility*-type definition. This is consistent with how Generator Owner/Operator and Generating Facility are treated, defined, and included in the registration criteria.

Data center sites can have a significant number of physical and electrical configurations. For example, many data center sites can consist of one or more buildings, with each building having one or more connections to the Bulk Power System at the same location. In some cases, data center owners may consider that each building is an individual data center, while in other cases the collection of buildings connected to the grid is viewed as the data center. The risk to the BES comes from the aggregate load behavior of facilities *at a single site*. This was emphasized in NERC’s “Characteristics and Risks of Emerging Large Loads” whitepaper that, while specific load types and thresholds to define Large Loads could not be defined across industry at that time, all agreed that any such definition should include “aggregation of load facilities at a single site behind one or more point(s) of interconnection that can pose reliability risks to the BPS.”

2. Replace CL Entity (CLE) with CL Operator (CLOP) and CL Owner (CLO)

The Trades recommend that the Computational Load Entity definition be replaced with new definitions that align with the “Owner” and “Operator” Model currently in place in the NERC Rules of Procedure for registration criteria. Today we have registration criteria for Transmission Owner (TO), Transmission Operator (TOP), Generator Owner (GO), and Generator Operator (GOP). This structure is in place because for Transmission and Generation facilities, there are often situations that occur where the owner of the facilities is different from the operator of the facilities. This registration criteria setup allows for the different entities to be responsible only for the activities that they are contractually setup for as the owner and/or operator.

For Computational Load Facilities, these facilities have the same type of structure. There are many cases where the owner is different from the operator. Therefore, because of the actual structure of the facilities and to align with existing NERC registration criteria, the

AZPS Comments on Proposed Revisions to NERC ROP: Computational Load Entity

May 8, 2026

Trades recommend replacing of the Computational Load Entity definition with Computational Load Owner and Computational Load Operator, with proposed definitions for each included below.

“Computational Load Owner” means the end-user entity that owns and maintains a Computational Load Facility, **and/or** the entity that owns and maintains a Computational Load Facility that hosts end-users that receives electric power **at a Computational Load Facility**.

“Computational Load Operator” means the end-user entity that operates a Computational Load Facility, **and/or** the entity that operates a Computational Load Facility that hosts end-users that receives electric power **at a Computational Load Facility**.

However, if NERC staff retains only one Registered entity, the Trades suggest the following modifications to the Computational Load Entity definition:

“Computational Load Entity” means the end-user **and/or** the entity that hosts end-users that receives electric power **for at a Computational Load Facility**.

The risk to the BPS and the gaps in NERC reliability standards, as noted in NERC’s “Assessment of Gaps in Existing Practices, Requirements, and Reliability Standards for Emerging Large Loads” is driven by Computational Load impacts and behavior, not *other* type of load, and Computational Load Entity should focus only on the amount of data-centric/information technology type of load.

3. Clarify the Computational Load (CL) definition:

The Trade Associations recommend that the proposed Computational Load definition focus on the power electronic-based information technology equipment, which is the focus of the risks to the BPS as documented by NERC’s Large Loads Task Force and Large Loads Working Group.

NERC’s “Assessment of Gaps in Existing Practices, Requirements, and Reliability Standards for Emerging Large Loads” identifies risks associated with the behavior and impacts of computational loads. Those risks arise from the data-centric and information technology characteristics of the load, not from unrelated load at the same site. The definition should therefore focus only on the amount of computational load and should not sweep in other types of loads that do not raise the same reliability concerns.

“Computational Load” means Load comprised **primarily** of electric power demand from **power electronic-based** information technology equipment.

AZPS Comments on Proposed Revisions to NERC ROP: Computational Load Entity

May 8, 2026

4. Either (a) increase Computational Load thresholds or (b) pursue an approach comparable to the Category 1/Category 2 designations for Generation.

a. NERC should increase its 20 MW and 60 kV Thresholds

The Trades recommend NERC reconsider the registration thresholds proposed in Appendix 5B. The proposed 20 MW/60 kV threshold with the 1MW Computational Load criteria is set unnecessarily low and sweeps in a substantial number of facilities that pose no material impact to the reliability of the Bulk-Power System and, thus, does not justify the burden of NERC registration. The current Rules of Procedure, Registration, and Standard Drafting efforts related to Computational Loads should focus on those loads that pose the most meaningful risk to the Bulk Power System. The threshold criteria (20 MW, 60 kV) is too low and we recommend the threshold be set to 75 MW connected at $\geq 100\text{kV}$ for Computational Load Facilities, consistent with the NERC Bulk Electric System definition for dispersed power producing resources (i.e., inverter-based resources) – where the current criteria is greater than 75 MVA at a common point of connection at $\geq 100\text{kV}$ is utilized.

Although NERC has developed a Technical Reasoning Proposed Computational Load document, it does not provide sufficient explanation or practical examples to justify the selected thresholds in the proposed registration criteria. The Trades are concerned that NERC's proposed approach could inadvertently create compliance obligations for Distribution Provider (DP) entities by effectively deeming them Transmission Owners (TO) or Transmission Operators (TOP). It is critical to ensure that DPs do not lose the radial system exemption when solely serving load and not operating transmission facilities. Clear justification and refinement of the proposed thresholds are required to demonstrate that there are identifiable and material risks to the BES.

In addition, NERC's "Characteristics and Risks of Emerging Large Loads" indicated that "most of the survey respondents qualified 'large' as greater than 50 MW, and the single size number most commonly suggested was 75 MW," and the NERC Alert 2 results confirmed this. NERC's "Assessment of Gaps in Existing Practices, Requirements, and Reliability Standards for Emerging Large Loads" refers back to the first Whitepaper and in the definition section, and focuses on really large loads and how several smaller loads (still much bigger than 20 MW) are not as big of a risk.

AZPS Comments on Proposed Revisions to NERC ROP: Computational Load Entity

May 8, 2026

“The potential impact of several 100 MW loads may be less than the impact of a 500 MW load and may require different treatment.”

Furthermore, the NERC “Ten benchmarks of an Effective Reliability Standard” white paper emphasizes that “each [NERC reliability standard] requirement is ‘not a lowest common denominator’ exercise but instead achieves an objective that is the best approach for bulk power system reliability.” The same applies to definitions and reliability criteria. NERC’s decision to select 20 MW appears to be a “lowest common denominator” choice, instead of a higher value that the majority of the industry agrees with.

The Trades recommend NERC adopt an initial threshold of 75 MW at a common point of connection of 100 kV or higher. This pairing adheres to the current Bulk Electric System definition and appropriately balances reliability risk to compliance burden. A 75 MW/100 kV starting point focuses computational load entity registration on the facilities most likely to have a material impact on Bulk-Power System reliability. It would also give NERC and the Regional Entities time to identify entities for registration, develop Reliability Standards, and gain compliance oversight experience with a manageable population. Further, it avoids imposing new obligations on facilities whose individual contribution to reliability risk is limited.

b. NERC should pursue a Category 1/Category 2 approach for Computational Loads

If a threshold lower than 75 MW is deemed necessary by NERC staff, the Trades recommend that NERC staff enhance the Technical Reasoning document to reflect the analysis/evidence demonstrating the lower facility threshold is needed for BPS reliability. In addition, in this scenario the Trades also recommend that NERC staff adopt the same registration criteria model that was developed for IBRs with the Category 1 GO/GOP and Category 2 GO/GOP. NERC should create a new Category 1 Computational Load Owner/Operator (CLO/CLOP) and Category 2 CLO/CLOP definitions, and structure the MW and voltage criteria for each to align with the Category 1 GO/GOP and Category 2 GO/GOP. The Category 1/Category 2 approach to address the different security requirements needed for each categories. To create this structure, the following definitions are proposed:

- **“Computational Load Facility”** means a collection of electrical equipment located at one or more buildings, structures, or installations that is designed to receive power, is served through electric facilities from one or more points

AZPS Comments on Proposed Revisions to NERC ROP: Computational Load Entity

May 8, 2026

of connection and meets the criteria in either Category 1 or Category 2 Computational Load Owner/Operator registration categories.

- “Category 1 CLO/CLOP” should be aligned with the BES criteria, which would be entities who own/operate Computational Load Facilities (CLFs) that are **≥ 75 MW** of Computational Load connected at **≥ 100 kV**.
- “Category 2 CLO/CLOP” should be aligned with the lower threshold value supported by NERC and industry, which would be entities who own/operator Computational Load Facilities that are **≥ 20 MW** of Computational Load connected at **≥ 60 kV**.

With NERC expected to address computational load risk on an accelerated timeline by the end of 2026, this structure would allow the entire ERO to focus its initial entity registration, standards development efforts, and compliance education activities on the highest-risk CLOs/CLOPs first. This framework also establishes a more manageable path to subsequently address lower-risk Category 2 CLOs/CLOPs in future work phases. Finally, this structure would enable a risk-based compliance approach by allowing more stringent reliability and security requirements to be applied to Category 1 CLOs/CLOPs compared to Category 2 CLOs/CLOPs.

5. Additional Comments

The draft Computational Load Entity registry criteria uses the word “capability” which can be interpreted as the maximum load of a Computational Load Facility (CLF). For example, if the CLF facility has a maximum possible load demand of 100 MW, then this is the Load “capability” of the CLF. However, this “capability” term as used with Load is likely not widely understood or utilized today. We recommend an alternative to the use of “capability” such as “**maximum peak demand**” of a Facility to be clearer. In addition, evaluating a CLF’s impact on the BES should not rely solely on its size. Incorporating operational characteristics such as interruptibility, load flexibility, and the availability of on-site or contracted backup generation will provide a more comprehensive and accurate assessment of system impacts.

Additionally, the proposed registration criteria includes reference to a “point of interconnection” at 60 kV. “Point of interconnection” is primarily a term referring to generation interconnections – interconnections that most often rely on one generating tie-

AZPS Comments on Proposed Revisions to NERC ROP: Computational Load Entity

May 8, 2026

line. Data centers often have many connections to the Bulk Power System at the same location. The ownership of equipment can also vary site to site, in which some connect to utilities directly at the high voltage kV bus by owning the high-voltage transformation, while others do not own the transformation or equipment greater than 60 kV.

In NERC's current proposed Computational Load Entity registration criteria, it states "contributes to an aggregate connected Load capability." How will NERC determine the amount of Computational Load or the amount of aggregate connected load thresholds? Will it be based on the "nameplate" of the various racks in a data center warehouse or some other methodology? Regardless of the method(s) for determining these values, the Trades recommend that this should be confirmed between the CLE and its Regional Entity and NERC should not be dependent on other registered entities for providing this information to Regional Entities. Critically this criteria states Load, not Computational Load. We believe this is incorrect and should be updated to state "Computational Load capability." In our proposed Computational Load Facility definition, we have included the use of Computational Load in the definition so that the focus is only on data-centric/information technology load. The Trades ask "How is a Computational Load fundamentally different from other large traditional loads such as steel mills, food processors, or automobile manufacturers? All large loads must be studied to ensure their interconnection and operation do not create grid reliability issues. There is concern that, as written, it may capture facilities that are not data centers or cryptocurrency mining operations, which are the business types that NERC has identified as potentially impacting the reliability of the BES. For example, many industrial and commercial facilities may operate more than 1 MW of information technology equipment (either now or in the future) such as servers, storage, and networking hardware to control and support processes, yet these are not solely or primarily computational in nature. In addition, it seems unreasonable that the loss of a 1 MW load, regardless of load type, would have a significant impact on the BES, therefore including a de minimis criteria of 1 MW of Computational Load drives significant confusion, pulls in a significant number of entities unnecessarily and has no justification pertaining to actual risk to the BPS. The 1 MW de minimis threshold conflicts with the material impact standard applied throughout Appendix 5B and would expand registration beyond entities capable of materially impacting BPS reliability. The 1 MW de minimis threshold should be removed from the Computational Load threshold.

To build on the concerns shared and the alternatives provided the Trades ask the following questions.

- "How will a Computational Load Entity(ies) be identified for registration under the proposed framework?"

AZPS Comments on Proposed Revisions to NERC ROP: Computational Load Entity

May 8, 2026

- Will the Transmission Owner (TO), Transmission Operator (TOP), or Transmission Planner (TP) be responsible for identifying this Load(s) and sharing that information with their Regional Entity?
- Additionally, will responses to the NERC Level 3 Alert on Computational Loads be used to inform or determine registration requirements?

In addressing these questions, the Trades recommend that the identification CLEs for potential registration should be the responsibility of the Regional Entity and NERC should not be dependent on other registered entities for providing this information to Regional Entities.

Appendix B

It is unclear on how large load facility topology/interconnection configurations may impact the applicability of the proposed registration criteria. Recommend that the NERC/ERO quickly develop a CMEP Practice Guide on this new registration criteria for review by industry and DT to ensure no unintended consequences from this new criteria.

The Trades also note that NERC's document "Summary of Revisions" states candidates for CL registration must meet all the criteria in the proposed revisions in Appendix 5B in order to be registered as a Computational Load Entity, however, that is not explicitly stated in the proposed changes in Appendix 5B and should be stated.

Bright-line Criteria

Computational Load Entity (CLE) registration should be based on clear, bright-line criteria coupled with a formal exception process similar to the BES definition framework. Such an approach would allow an entity to demonstrate, through documented technical justification, that its facilities do not have a material impact on the Bulk Electric System (BES). This balance is necessary to ensure that registration and compliance obligations are applied only where there is a substantiated reliability risk, while avoiding unnecessary regulatory burden for entities whose operations do not pose a material threat to BES reliability.

NERC ROP Revisions: Computational Loads and Computational Load Entity – BPA Comments

Appendix 5B

BPA understands NERC's ROP revisions proposal has the following registration qualifiers for a Computational Load Entity:

- 1) contributes to an aggregate connected Load capability greater than or equal to 20 MW;
- 2) at a single point of interconnection to the Bulk Power System at a voltage greater than or equal to 60 kV; and
- 3) hosts 1 MW or greater of Computational Load.

In BPA's experience, data center interconnections vary.

Many large data centers interconnect at High Voltage (HV) level, 230-kV or 115-kV, and own step-down transformers and Medium Voltage (MV) equipment, usually 34.5-kV and 12.5-kV). BPA believes these will clearly fit into the proposed NERC ROP registration criteria.

However, BPA also sees large data centers that interconnect at the MV level, as illustrated below for a 200 MW data center. BPA understands that an interconnecting utility will own and operate HV to MV step-down transformers and MV substation and the change of ownership between a utility and a data center occurs at 34.5-kV level. BPA recommends the definition clarify that the described interconnection meets the NERC definition.

To achieve BPA's recommendation regarding the Computational Load Entity (CLE) definition, BPA suggests revising the language listed below:

III(c) Computational Load Entity:

III.c.1 Computational Load Entity that encompasses all of the following:

- 1) **One or more facilities at a single site that** contributes to an aggregate connected Load capability greater than or equal to 20 MW;
- 2) ~~at a single point of interconnection to the Bulk Power System at a voltage greater than or equal to 60 kV; and~~
Connects to the Bulk Power System (BPS) operated at a voltage greater than or equal to 60kV on the high side and behind one or more common Points of Interconnection (POIs) or Service Delivery Points; and

BPA believes 3MW is appropriate and should be the lowest MW value, as this is currently identified for AGC. BPA understands this to be the threshold for modeling and operational monitoring. BPA recommends revising the language to:

- 3) hosts ~~1~~ **3**MW or greater of Computational Load.

Appendix 2:

Accompanied by BPA proposed revisions for Computational Load Entity, BPA recommends revising the BES definition to include 'Computational Loads'.

BES Definition:

"Bulk Electric System" or "BES" means unless modified by the lists shown below, all Transmission Elements operated at 100 kV or higher, [Computational Loads](#), and Real Power and Reactive Power resources connected at 100 kV or higher. This does not include facilities used in the local distribution of electric energy.

BPA recommends adding a new 'Inclusion', proposed I6.

Inclusions:

I6 - [Computational Loads, and the associated system\(s\) that connects it to the BES network for a Computational Load Entity.](#)

Exclusions:

BPA recommends the following 'Exclusions' language revisions to accompany BPA's recommended I6 'Inclusion' language above.

- **E1 - Radial systems:** A group of contiguous transmission Elements that emanates from a single point of connection of 100 kV or higher and:
 - d) Only serves Load, [not identified in Inclusion I6](#). Or,
 - e) Only includes generation resources, not identified in Inclusions I2, I3, or I4, with an aggregate capacity less than or equal to 75 MVA (gross nameplate rating). Or,
 - f) Where the radial system serves Load, [not identified in Inclusion I6](#), and includes generation resources, not identified in Inclusions I2, I3 or I4, with an aggregate capacity of non-retail generation less than or equal to 75 MVA (gross nameplate rating).

- **E3 - Local networks (LN):** A group of contiguous transmission Elements operated at less than 300 kV that distribute power to Load, [not identified in Inclusion I6](#), rather than transfer bulk power across the interconnected system. LN's emanate from multiple points of connection at 100 kV or higher to improve the level of service to retail customers and not to accommodate bulk power transfer across the interconnected system. The LN is characterized by all of the following:

BPA proposes a new bullet 'd' in 'E3' to not allow exclusions for Computational Loads, using the same concept as bullet 'a' for connected Generation.

d) Limits on connected loads: The LN and its underlying Elements do not include Computational Loads identified in Inclusions I6.

BPA appreciates the compressed timeline NERC working under to address reliability risks posed by computational loads, and the necessary focus on the most critical actions to take in 2026 to address them. If this means the BES definition revisions are not incorporated now, BPA requests NERC open the BES definition for revision in early 2027 to account for Computational Loads and other changes to the nature of the BES that may warrant revision to its definition. This will address the risk that a CLE won't have BES equipment using the existing BES definition. BPA believes BES definition clarity is an important step so the standard(s) being developed by Project 2026-02 are capturing the correct BES equipment before their implementation phase.

CenterPoint Energy Houston Electric, LLC (CEHE) is grateful for the opportunity to review and comment on the NERC proposed revisions to the NERC Rules of Procedure (ROP) to integrate Computational Loads as a registered Entity class.

CEHE is supportive of the attached comments submitted by the Trade Associations to NERC. The Trade Associations are the: American Public Power Association (APPA), Edison Electric Institute (EEI), Large Public Power Council (LPPC), National Rural Electric Cooperative Association (NRECA), and the Transmission Access Policy Study Group (TAPS).

CEHE looks forward to continuing to collaborate with NERC via the Project 2026-02 Computational Loads Standard Drafting Team and the NERC LLWG on these ongoing efforts to design risk-based reliability requirements for Computational Loads.

Thank you.



Leo McAfee Jr.
Senior NERC Policy and Compliance Consultant

CenterPoint Energy | CEHE Policy and Compliance
CenterPointEnergy.com

In addition to supporting the comments submitted by APPA, SVP provides the following comments on the Computational Loads SAR and proposed ROP revisions which include a new definition for “Computational Load Entity” and registration criteria for CLEs.

- **COMPUTATIONAL LOAD ENTITY “CLE” REGISTRATION**

SVP agrees that registration of Computational Load Entities is necessary to help ensure that interconnecting entities commit to meeting baseline reliability expectations, and SVP supports the registration criteria proposed in the Rules of Procedure, Appendix 5b **[1)load capability greater than or equal to 20M; 2) at a single point of interconnection to the Bulk Power System at a voltage greater than or equal to 60kV; and 3) hosts 1MW or greater of Computational Load.]**

- **CLE DEFINITION**

SVP considers the proposed definition of **Computational Load Entity**, “the end user or the entity that hosts end-users that receives power for Computational Load,” to be unnecessarily broad and more likely to result in conflicting accountability and strain any coordination efforts between TOPs and the CLEs within their system. SVP recommends that the Computational Load Entity be identified as the host/facility owner; this will streamline any coordination or Interpersonal Communication requirements that evolve from the Large Loads Bridge Standard project. (e.g.: Level 3 NERC Alert, Essential Action #7 – “TOPs, RCs, and BAs should establish Interpersonal Communication capabilities with computational loads to improve their situational awareness and joint operating procedures to ensure the reliability operation of the BPS. . .”)

- **SCOPE of STANDARD AUTHORIZATION REQUEST**

While the purpose of the Standard Authorization request is generally understood by the industry, the SAR itself lacks enough specificity about the reliability risks it intends to address. SVP recommends that the SAR identify the risks the Essential Actions are meant to mitigate, using the “Essential Actions” in the May 4th Level 3 NERC Alert pertaining to Computational Loads as guidance. This will assist both the drafting team and voting entities determine whether the initial bridge standard meets the purpose of the SAR.

Sincere Regards,

Val

Valentina Guzman Ridad

Division Manager, Compliance | Electric Department
881 Martin Avenue | Santa Clara, CA 95050



Concern Regarding 1 MW Minimum Computational Threshold

The proposed 1 MW computational load threshold may be too low to reliably indicate the presence of system-relevant computational behavior. A facility with 1 MW of computational load within a 20MW aggregated load is unlikely to exhibit the characteristics that NERC is intending to address. We suggest evaluating whether a higher threshold or ratio-based threshold would better reflect reliability risk.

SPP recommend NERC evaluate whether:

- A higher computational load minimum (for example, 3–5 MW) better reflects the threshold at which computational characteristics materially impact system stability, or
- Additional technical justifications are provided to support the 1 MW cutoff.

Need Precise Definition of “Point of Interconnection at ≥ 60 kV”

We request clarification on how the point of interconnection at 60 kV or above is to be determined. This is critical for distinguishing true transmission-connected computational loads from distribution-fed commercial or mixed-use facilities. A defined measurement methodology would help ensure consistent registration and audit application.

Alan Wahlstrom N.C.S.O
Lead Engineer
Reliability Standards Development Group
Southwest Power Pool

To whom it may concern,

Please find the following suggested change for Appendix 5B - Statement of Compliance Registry Criteria - Revision 9:

Registration Criteria:
III(c) Computational Load Entity:
III.c.1 Computational Load Entity that 1) contributes to an aggregate connected Load capability greater than or equal to 20 MW of Computational Load; and 2) at a single point of interconnection to the Bulk Power System at a voltage greater than or equal to 60 kV; ~~and 3) hosts 1 MW or greater of Computational Load.~~

The reasoning for this is that with the currently proposed definition (hosting 1MW of greater of Computational Load), a large number of facilities may be pulled into the registration criteria that are not AI data centers. 1MW of computational load may inadvertently require office buildings, university campuses, research centers, etc. to register as a Computational Load Entity.

By amending the first item to focus on the required amount of computational load, and not an aggregate load plus a relatively small portion of computational load, the definition becomes more salient and succinct.

Thank you,

Eric Mannarino
Global Design & Construction
Design Delivery Principal, Electrical Engineering

EQUINIX | 1 Lagoon Dr, Redwood City, CA 94065

EQUINIX 	
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Sylvia and/or NERC Representative Responsible for Obtaining Industry Comments,

I am submitting comments for a for all 4 sectors 1,3,5, and 6 for the Computational Load Entity ROP Revisions solicitation.

Dominion Energy supports EEI comments with no additional comments.

Matt Bullard

NERC Reliability Compliance Consultant

Dominion Energy

Corscale Data Centers agrees with the comments submitted by the Data Center Coalition.

Best Regards,

Dan Vanderplas

Data Center Campus Director



13760 University Blvd.
Gainesville, VA 20155

CORSCALE.COM

Dear NERC,

Corscale Data Centers agrees with and support the comments submitted by the Data Center Coalition.

Regards,

Ludo Hintos, PE

Vice President, Infrastructure Planning & Development



13760 University Blvd.
Gainesville, VA 20155

CORSSCALE.COM

To whom it may

concern,

I am writing to provide comments on the proposed updates to the Rules of Procedure regarding the inclusion of Computational Load Entities.

The initiative to address risks associated with large and rapidly scaling computational loads is both timely and necessary. As these loads increasingly influence system planning and reliability considerations, establishing a clear framework is critical.

One consideration that may be worth incorporating into both the ROP and related standards discussions is how underlying demand is characterized.

Not all large computational loads are equally durable. A portion of projected demand may be conditional on evolving factors such as:

compute architecture and efficiency improvements
siting and co-location decisions
operational strategies and workload distribution
underlying technology changes

As a result, treating all projected load as structurally fixed may introduce risk into planning and standard-setting processes.

In particular, there may be value in distinguishing between:

structural (persistent) load
conditional or transient load
load that may be reduced or eliminated through technological or operational changes

Incorporating this type of differentiation could help ensure that standards are aligned not only with current demand projections, but also with how those demands may evolve over time.

This consideration has implications for:

long-term reliability planning
infrastructure investment efficiency
risk assessment associated with large load integration

Appreciate the opportunity to provide input on this important topic.

Best regards,
Neil P. Osnato

Neil P. Osnato
Founder

Persistence Analytics Group LLC

Regarding the Rules of Procedure revisions relative to Computational Load (comments due May 15), the Large Public Power Council (LPPC) supports the comments submitted by the Trade Associations. Thanks.

Joe McClung

JEA Electric Compliance Program Manager

Dear NERC,

After reivew Corscale Data Centers agrees with and support the comments submitted by the Data Center Coalition.

Thank you,

Tyler Kent, P.E.

Project Director, Infrastructure Delivery



9830 Colonnade Blvd., Suite 405
San Antonio, TX 78230

CORSCALE.COM

May 12, 2026 | Via Electronic Filing: ROPcomments@nerc.net

Comments of Crusoe Technologies LLC

NERC Rules of Procedure Revision — Computational Load Entity Registration Criteria (Appendix 5B)

45-Day Comment Period | April 1 – May 15, 2026 | Project 2026-02

I. Introduction

Crusoe Technologies LLC ("Crusoe") submits these comments on NERC's proposed Appendix 5B registration criteria for a new functional entity type — the Computational Load Entity (CLE). Crusoe operates large-scale artificial intelligence compute infrastructure interconnected to the bulk power system (BPS) and participates actively in ERCOT's Batch Zero large load interconnection proceeding.

Crusoe supports the CLE framework, with more specific definitions. NERC's May 4, 2026 Level 3 Alert documented widespread, simultaneous load reductions exceeding 1,000 MW from computational facilities — events that create real BPS stability exposure. A registration and compliance framework with mandatory Reliability Standards is the right response. That said, any definitions and registration criteria should be grounded in load size and grid impact, applicable to all large load facilities regardless of industry type — not targeted at a specific end use of electricity. Crusoe's comments address two specific issues: the 20 MW threshold is too low and will not accomplish what NERC intends, and the proposed definitions are overbroad and require refinement.

II. The 20 MW Threshold Is Too Low — 75 MW Is the Right Number

The proposed criteria require registration where a facility contributes to aggregate connected load capability of ≥ 20 MW at ≥ 60 kV and hosts ≥ 1 MW of computational load. The 20 MW floor does not reflect large load impacts on the BPS.

A 20 MW load tripping offline — even instantly — creates an immeasurable frequency deviation on the Eastern or Western Interconnection. Automatic generation control and governor response absorb events of this size routinely without any reliability consequence. NERC's own Level 3 Alert identified the reliability problem as large-scale, simultaneous drops from clusters of computational facilities — not from individual 20 MW sites. Registering 20 MW facilities does not solve that problem. It creates an enormous administrative burden for the wrong population while diluting NERC's enforcement focus.

ERCOT — the RTO with the largest concentration of emerging computational load in North America — set its large load interconnection threshold at **75 MW**. This threshold reflects decades of ERCOT operational experience with the load size at which a single facility begins to affect transmission planning assumptions, SCED constraint dispatch, and reliability coordination. Crusoe respectfully urges NERC to adopt 75 MW as the CLE threshold and to explain, with technical justification, if it believes a different number better reflects BPS material impact.

III. The 1 MW Computational Load Sub-Criterion Sweeps In the Wrong Facilities

The proposed criteria require only that a qualifying facility 'hosts 1 MW or greater of Computational Load' — defined as load from servers, storage, and networking hardware. This definition is overly broad: it sweeps in hospitals, universities, corporate offices with modest server rooms, and industrial facilities like steel mills with incidental on-site IT equipment. None of these are the purpose-built, utility-scale infrastructure that prompted NERC's Level 3 Alert, and imposing full NERC compliance obligations on incidental IT users bears no relationship to the reliability risk NERC is managing

That is not the population that tripped 1,000 MW simultaneously and prompted NERC's Level 3 Alert. The Level 3 Alert, the Level 2 Alert before it, and the Large Loads Working Group white paper all focused on purpose-built, utility-scale AI and data center infrastructure. Imposing NERC Reliability Standards compliance — with audit obligations and civil penalty exposure up to \$1 million per violation per day under FPA Section 215 — on incidental IT users bears no relationship to the reliability risk NERC is managing. Crusoe recommends that the computational load sub-criterion be set at 50 MW of dedicated computational load, or alternatively that computational load constitute at least 50% of the facility's aggregate connected load capability. Either formulation limits registration to purpose-built computational infrastructure — the actual source of BPS risk — and excludes conventional commercial and industrial facilities with incidental IT. As such, Crusoe recommends refining the definition of Computational Load to capture only loads that meet the MW threshold and are "predominantly comprised of" or where "the majority of electric power demand" comes from IT equipment.

IV. Proposed Revision to Appendix 5B Criterion 2

Current Language:

A Computational Load Entity that (1) contributes to an aggregate connected Load capability greater than or equal to 20 MW; (2) at a single point of interconnection to the BPS at a voltage ≥ 60 kV; and (3) hosts 1 MW or greater of Computational Load.

Crusoe's Recommended Language:

A Computational Load Entity that (1) contributes to an aggregate connected Load capability greater than or equal to 75 MW; (2) at a single point of interconnection to the BPS at a voltage ≥ 100 kV; and (3) hosts 50 MW or greater of Computational Load [alternative: for which Computational Load constitutes at least 50% of aggregate connected Load capability at that point of interconnection].

Crusoe also recommends a 100kv voltage threshold instead of the proposed 60kv. Transmission-level interconnection is a meaningful indicator of BPS exposure and should remain a necessary condition..

V. One Additional Issue: Co-Located Generation and Storage

The proposed CLE definition does not address facilities that combine computational load with co-located generation or energy storage — the BYOG (Bring Your Own Generation) model increasingly common in large AI infrastructure deployments. A facility that must register as a CLE, a Generator Owner, and an Energy Storage Resource Owner may face multi-entity compliance obligations that

have not been reconciled. Crusoe requests that NERC issue guidance on multi-functional entity registration for co-located configurations before the CLE criteria are finalized.

VI. Additional Recommendations

Crusoe supports the comments submitted by the Data Center Coalition (DCC). As an additional recommendation if NERC maintains the 20 MW/60kV thresholds, NERC should adopt the same registration criteria model that was developed for Inverter-Based Resources with the Category 1 and Category 2 GO/GOP definitions and criteria. This would enable NERC and the industry to focus initially on reliability issues created by the larger Category 1 Computational Load Owner/Computational Load Operator categories.

Category 1 CLO/CLOP: Computational Load Facilities that are ≥ 75 MW of maximum peak demand of Computational Load connected at ≥ 100 kV

Category 2 CLO/CLOP: Computational Load Facilities that are ≥ 20 MW of maximum peak demand of Computational Load connected at ≥ 60 kV

VI. Summary of Recommendations

1. Raise the aggregate connected Load capability threshold from 20 MW to 75 MW.
2. Raise the Computational Load hosting sub-criterion from 1 MW to 50 MW — or require Computational Load to constitute at least 50% of aggregate connected load capability.
3. Raise the 60 kV voltage threshold to 100kV
4. Issue guidance on multi-functional entity registration for co-located generation and storage configurations before the CLE criteria are finalized.

Crusoe thanks NERC for its leadership in addressing the BPS reliability implications of large computational loads. The Level 3 Alert demonstrates that the urgency is real. Getting the threshold right — so that mandatory compliance obligations fall on the facilities that actually affect BPS reliability — is essential to making the CLE framework effective. Crusoe is available to provide additional technical input if that would assist NERC's deliberations.

Respectfully submitted,

Crusoe Technologies LLC

Contact: Bharath Ravulapati

May 15th, 2026

Filed electronically to: ROPcomments@nerc.net



Notice of Proposed Revisions to the NERC Rules of Procedure and Request for Comments – Computational Load Entity

NERC Rules of Procedure Appendix 2 (Definitions Used in the Rules of Procedure); Appendix 5A (Organization Registration and Certification Manual); and Appendix 5B (Statement of Compliance Registry Criteria)

[Proposed Changes to Rules of Procedure](#)

[Appendix 2 Clean](#) | [Redline](#)

[Appendix 5A Clean](#) | [Redline](#)

[Appendix 5B Clean](#) | [Redline](#)

[Summary of Changes](#)

Proposed Revisions to Rules of Procedure – NERC is proposing revisions to its Rules of Procedure to develop a new registered entity type, called Computational Load Entity. NERC is posting the proposed revisions to NERC Rules of Procedure Appendix 2, Appendix 5A, and Appendix 5B for public comment for 45 days from **April 1, 2026** to **May 15, 2026**. Comments must be submitted electronically to

ROPcomments@nerc.net.

The proposed revisions address recommendations from the NERC Large Loads Working Group (a subgroup of the NERC Reliability and Security Technical Committee) whitepaper “[Assessment of Gaps in Existing Practices, Requirements, and Reliability Standards for Emerging Large Loads](#)” and are further supported by the findings in the “[NERC Aggregated Report on NERC Level 2 Industry Recommendation: Large Load Interconnection, Study, Commissioning, and Operations](#)” analyzing responses to a [Level 2 Alert](#) on the interconnection of large loads. The recommendations and findings indicate a need for a new registered entity to comply with Reliability Standards (currently under development in [Project 2026-02 – Computational Loads](#)) to reduce risks to the reliability of the Bulk Power System.

In pursuing its mission to assure the effective and efficient reduction of risks to the reliability and security of the grid, NERC works with owners, operators, users, and consumers as well as federal, state, and provincial regulators across North America. To that end, NERC welcomes input from all stakeholders on the proposed revisions and values the opportunity to engage with stakeholders on how the proposed revisions can contribute to the reliability of the Bulk Power System.

For more information or assistance, please contact [Marisa Hecht](#) via email or at **404-446-9620**.

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The Data Center Coalition is pleased to submit these comments on NERC's proposed Rules of Procedure definitions and registration criteria for Computational Loads.

Like all customers, the data center industry depends on a reliable electricity grid in order to fulfill its role as a vital backbone of our modern economy. In turn, data centers offer a number of reliability benefits to the grid, including significant investments in new energy resources and next-generation technologies. As recently demonstrated during Winter Storm Fern, data centers can also serve as grid assets that help promote reliability during energy emergencies caused by extreme weather.

It is critical to ensure proposed definitions and registration criteria by NERC are based on complete evidence, the technical capabilities and specifications of data center equipment, and do not disregard important developments or information from key markets. In Virginia, improved communication between NERC, regional entities, utilities, and data center operators have greatly reduced — if not altogether eliminated in certain instances — load transfer events in Virginia. In Texas, DCC is unaware of any load transfer events that have been caused by data centers, based on ERCOT's reporting. This type of close collaboration, communication, and technical information sharing is essential to the upcoming work with NERC.

While NERC has elected to pursue a pathway that conflates and singles out "computational loads," we emphasize that any new definitions and registration criteria should be defined by any type of large load facility – and the corresponding load size, voltage level, and operational characteristics of any such large load facility - not simply by industry type or end use of electricity. With that fundamental concern noted, DCC is committed to leaning in as an engaged partner with NERC and transmission entities to help ensure a reliable grid for all customers and offers the following comments. In addition, while the DCC understands the urgency of the NERC computational load work, the recently released NERC Level 3 Alert on Computational Load Modeling, Studies, Instrumentation, Commissioning, Operations, Protection, and Control is an important tool that should help get essential actions in place by the electric transmission industry. Given the scale and criticality of this Level 3 Alert, it would be prudent to wait for the industry response to this Level 3 Alert and corresponding essential action implementation before moving forward with the registration criteria and standard development for Computational Loads. The DCC also urges NERC to publicly release its legal analysis that NERC stated was ongoing throughout 2025, which we understand is supporting the conclusion that NERC has the authority under Section 215 of the Federal Power Act to register Computational Load Entities and impose mandatory reliability standards on them. This is a significant expansion of NERC's jurisdictional reach, and the data center industry, other computational load industries, and the electric industry all deserve to understand the legal reasoning before any registration framework or standard is advanced. Publishing this analysis would allow for informed stakeholder engagement, support a more defensible rulemaking record, and ensure that the process is lawful and reflects the transparency and due process the industry expects from NERC.

- Summary of RoP Changes

- NERC has decided to target large computational loads, which appears to specifically exempt other types of large loads that also have a material impact on the bulk power system. Particularly with the relatively low thresholds proposed, it is unclear how a 20 MW data center would have a substantially greater impact to the BPS than, say, a multi-hundred MW mining load or manufacturing facility.
- Appendix 2
 - The definition for “Computational Load” is rather generic and may be difficult to implement and enforce due to the ambiguities it contains. Specifically, the terms say that “Load comprised of electric power demand from IT equipment...” Thus, if ANY portion of the load is IT-based equipment, then this would technically constitute as a Computational Load. For some examples:
 - A steel mill likely has some servers or computers on site that ultimately are IT equipment; however, the vast majority of the load is an industrial process and not intended to be captured by this definition.
 - Hospitals, large office buildings, colleges/universities are additional examples that all are large loads that will have plenty of IT equipment, including servers, storage systems, and networking equipment. But all of these facilities should not be intended to be captured by this definition.

Therefore, NERC should consider some improvements to the definition to scope and refine it. One example would be to state “Load primarily comprised of electric power demand from IT...” or “Load where the majority of electric power demand from IT...”. This captures the same intent but provides some clear specifications on how to apply it.

- NERC should consider whether the “Computational Load Entity” (“CLE”) is sufficient for the purposes of future standards applicability and registration. Does this refer to the owner or operator of the facility or the owner and operator of the computational load? Which is the main target of standards? What if the owner and operator are different entities, which is a common structure in the data center industry?
- NERC may want to further clarify what “end-user or the entity that hosts end-users” means exactly, with some examples to support registration efforts. The industry generally believes this refers to, for example, data center locations that may have multiple tenants, but examples/confirmation from NEC on this is critical.
- Appendix 5B

- The criteria for CLE is generally missing the overall risks stated by NERC and industry in the various reports, as written. The criteria essentially states that a 50 MW load that has *only* 1 MW of Computational Load would then need to be registered as a CLE. A 1 MW threshold for registration and applicability to NERC standards seems very misaligned with the findings in NERC reports. We recommend that the 1 MW threshold requirement be removed entirely from the definition.
- In addition, a more straightforward definition would be: “contributes to aggregate connected **Computational** Load greater than 20 MW at a single point of interconnection to the BPS at a voltage greater than or equal to 60 kV.” It is unclear why a definition such as something like this couldn’t achieve the same outcome and be more aligned with other NERC registration efforts (e.g., Category 2 GO/GOPs) and general understanding of BPS impact/risks.

With those general comments provided, DCC would like to propose the following modifications to the proposed definitions and registration criteria, while also providing new definitions that provide better alignment with existing NERC registration criteria and definitions.

- Modify the new term, “Computational Load”, to reference this load being primarily made up of power-electronic-based information technology equipment and remove any unnecessary examples of that equipment.
 - **“Computational Load”**: Load comprised **primarily** of electric power demand from **power electronic-based** information technology equipment
- Create a new definition, “**Computational Load Facility**”, to align with how existing NERC standards apply to Facilities (not entities) and include the updated registration criteria in this new definition. The registration criteria include updated MW threshold to 75 MW, updated voltage connection to 100kV, and changes the word “capability” to “maximum peak demand.”
 - **Computational Load Facility (“CLF”)**: a collection of electrical equipment located at one or more buildings, structures, or installations that is:
 - (1) designed to receive power
 - (2) has an aggregate connected maximum peak demand of Computational Load greater than or equal to **100 MW**
 - (3) and is served through electric facilities at a voltage level greater than or equal to **100kV** from a single point of connection.
- Modify the new term, “Computational Load Entity”, to reference the CLF definition and add some clarifying edits to the definition:

- **“Computational Load Entity”**: the end-user **and/or** the entity that hosts end-users that receives electric power **at a Computational Load Facility**.
- Ideally, we would also like to propose that the Computational Load Entity definition turn into two separate Owner/Operator definitions, to align with the existing NERC registration criteria/definitions for Transmission Owners, Transmission Operators, Generation Owners, and Generation Operators. Updating CLE to CLO/CLOP is necessary as many data center facilities today have different owners and operators, just as occurs with Transmission and Generation facilities.
 - **“Computational Load Owner” (CLO)**: the end-user entity that **owns and maintains** a Computational Load Facility, and/or the entity that **owns** a Computational Load Facility that hosts end-users that receives electric power at a Computational Load Facility.
 - **“Computational Load Operator” (CLOP)**: the end-user entity that **operates** a Computational Load Facility, and/or the entity that **operates** a Computational Load Facility that hosts end-users that receives electric power at a Computational Load Facility.
- If NERC insists on keeping the 20 MW threshold of Computational Load, we ask that NERC provide a detailed explanation and evidence why such a low threshold is needed, as NERC’s own reports and white papers do not align with this low threshold.
 - NERC’s Large Load Task Force (LLTF) White Paper #1 (Characteristics and Risks of Emerging Large Loads) states that most survey respondents listed large loads to be in the range of at least 50 to 75 MW, and data center facility size is generally increasing.
 - NERC’s LLTF White Paper #2 (Assessment of Gaps in Existing Practices, Requirements, and Reliability Standards for Emerging Large Loads) states that “the potential impact of several 100 MW loads may be less than the impact of a 500 MW load and may require different treatment.”
 - NERC’s report on the Level 2 Alert responses (“Aggregated Report on NERC Level 2 Industry Recommendation: Large Load Interconnection, Study, Commissioning, and Operations”) also includes a figure of the aggregate responses to question on what is the current, or proposed, MW threshold for respondent’s large load definition. The vast majority of responses were 50 MW or 75 MW. The report also stated that “the reporting [Distribution Providers] had thresholds typically below 20 MW.”
 - All these reports and recommendations lead us to recommending that NERC increase the threshold to 100 MW connected at 100kV for the proposed Computational Load Facilities.

If NERC should continue to insist on a lower threshold being needed, we recommend that NERC adopt the same registration criteria model that was developed for Inverter-Based Resources with the Category 1 and Category 2 GO/GOP definitions and criteria. This would allow the industry to first focus our standard development efforts on the larger Category 1 CLO/CLOP Facilities, and then in time address the lower Category 2 CLO/CLOP Facilities. Below are the proposed Category 1 and Category 2 CLO/CLOP definitions that align with the current Category 1 and Category 2 GO/GOP thresholds/criteria.

- **Category 1 CLO/CLOP:**
 - Computational Load Facilities that are **≥ 100 MW of maximum peak demand of Computational Load** connected at **≥ 100 kV**
- **Category 2 CLO/CLOP:**
 - Computational Load Facilities that are **≥ 20 MW of maximum peak demand of Computational Load** connected at **≥ 60 kV**

There are two primary types of data centers relevant to this discussion: (1) self-perform and enterprise data centers, in which the business owns and controls its servers and peripheral equipment and may or may not own the physical facility; and (2) multitenant and build-to-suit data centers, in which the facility owner leases space to one or more tenants, potentially encompassing both critical and non-critical load behind the same meter, where those tenants bear direct responsibility for the computational load activity occurring within their leased space. Reference the CIP-002-5.1a ERO Endorsed Implementation Guidance document that also defines Tenants and considers Tenants in the aspects of the standard.

Any attempt by NERC to define Computational Loads and Computational Entities must be cognizant of the complex and varied operational structures, agreements, contracts, and lease arrangements that currently exist between the owners, operators, and tenants of computational load facilities. These relationships were established under existing regulatory frameworks and commercial constructs that did not anticipate the definitional boundaries now being proposed. Accordingly, it is essential that NERC incorporate a clearly defined exception process to accommodate computational load facilities that were designed, contracted, and operationalized prior to the effective date of these rule changes; facilities that, through no fault of their own, may not align with new definitions or satisfy new requirements without significant operational or contractual disruptions and substantial retrofit costs. A blanket application of new definitions and requirements without such a process risks penalizing facilities for non-compliance with standards they had no reasonable basis to anticipate and could create unintended consequences for grid reliability if affected entities are forced into reactive operational changes.

Thank you for providing an opportunity to comment on the proposed Computational Load Entity Registry Criteria.

NERC should use the information it collects to provide greater transparency regarding large computational loads to the wider electricity market. Currently, market participants are struggling to understand where new large computational loads will interconnect, their size, and basic operational characteristics. Without this information, project developers, generators, power marketers and other electricity industry participants cannot forecast and plan for future grid conditions accurately and efficiently.

Large computational loads affect grid operations and locational electricity prices to a similar extent as generation resources of equivalent size. Yet, there is far less information available regarding large computational loads than there is for generation resources. For example, the Energy Information Administration publishes a public data base with basic information on the interconnection location and characteristics of generation resources (i.e., Form EIA-860 Data Base), but does not maintain an equivalent data base for large computational loads.

Private actors across the industry cannot make efficient investments in transmission and generation to serve the extraordinary demand forecasted from large computational loads without greater transparency, stifling investment and, ultimately, jeopardizing reliability. Electricity market participants should have access to certain information regarding each large computational load via a database of approved loads, their sizes, and the locations at which they plan to interconnect with the grid. In addition, market participants should have information regarding the portion of the load that will be firm or in another category. NERC should also provide basic information on the type of load, for instance, whether the studied load is for a data center or crypto mining, and whether the load will include new or existing co-located generation.

Justin R. Cockrell
Assistant General Counsel
DC Energy Management, LP

DPN Comments on NERC Proposed Rules of Procedure Revision:

Computational Load Entities (CLEs)

As NERC considers revisions to the Rules of Procedure Appendix 2 (Definitions Used in the Rules of Procedure) to add new Computational Load Entities (CLEs), the Digital Power Network (DPN) offers the following comments and recommendations to ensure the continued operational viability of large, flexible computing loads as the standards are drafted.

DPN is the largest coalition of Bitcoin miners and digital infrastructure providers, representing facilities with a significant footprint in the grid and energy sector. Bitcoin mining facilities and data centers are increasingly critical components of modern infrastructure that support financing buildout and providing demand flexibility. Any standards framework should therefore balance reliability objectives with the need to support the safe, secure, and economically viable operation of these facilities.

I. Appendix 2 (Definitions Used in the Rules of Procedure) Comments

Ensure Clear Statutory Authority

NERC should clearly articulate the statutory basis for creating the Computational Load Entity (CLE) category and the proposed definition to ensure the framework remains legally grounded.

While Section 215 of the Federal Power Act (FPA) authorizes NERC to develop mandatory reliability standards for the Bulk-Power System (BPS), subject to FERC approval, that authority is not based on the identity of a particular industry or type of end-user. Rather, it is based on whether an entity materially affects the reliable operation of the BPS.

Accordingly, NERC should continue to focus on functional reliability impacts rather than establishing a fixed, industry-specific category of “Computational Load Entity.” Historically, NERC registration and compliance obligations have been tied to reliability functions and material impacts on the BPS, rather than a facility’s status as a particular type of electricity consumer.

Further, whether a computational load materially impacts BPS, reliability may need to be evaluated on a case-by-case basis given regional and operational differences. At the same

time, other non-CLE load entities may also materially affect BPS reliability but would not fall within the proposed definition.

NERC should also clarify its statutory basis for imposing direct reliability standards on “end-users” of the BPS. Section 215 applies to “users, owners, and operators” of the BPS, but NERC has historically applied registration and compliance obligations based on an entity’s reliability function and material impact on BPS operations, rather than solely on its status as an electricity consumer.

Historically, crypto miners and data centers have generally been treated as end-users of electricity rather than entities directly subject to mandatory reliability standards. While NERC and applicable reliability entities may exercise indirect oversight through transmission operators, planners, balancing authorities, interconnection processes, and operating requirements, these facilities have not traditionally been subject to direct compliance obligations under the FPA solely by virtue of being large electricity consumers.

Further, NERC’s jurisdiction extends only to entities and activities affecting the BPS and does not extend to purely local distribution facilities or other systems outside the statutory definition of the BPS.

While DPN supports NERC’s mission to reduce risks to BPS reliability, expanding reliability standards through the BPS to directly regulate end-users raises broader legal and practical concerns that warrant careful consideration.

II. Foundational Principles for Initial Standards

Prioritize Coordination Before Compliance Requirements

The current standards drafting process appropriately emphasizes process-based standards. It is important that NERC promote coordination between large loads and relevant authorities.

Early and structured coordination will:

- Ensure stakeholder input is incorporated into standards development
- Reduce the risk of infeasible or misaligned requirements
- Improve overall implementation outcomes

Large loads should be active participants in the standards-setting process, not only subjects of compliance.

Streamline and Standardize Data Requests

As standards evolve, multiple entities (e.g., Transmission Planners, Planning Coordinators, Reliability Coordinators) may request similar data from large loads.

To avoid duplicative and burdensome requirements:

- Data requests should be standardized and streamlined across entities
- A coordinated framework should minimize repetitive submissions
- A lead authority or centralized process should be designated when possible

Protect Proprietary and Sensitive Information

Data centers and Bitcoin mining facilities frequently handle proprietary or third-party data. Energy usage and operational characteristics are integral to their business models and may be competitively sensitive.

NERC should incorporate clear data protection standards, including:

- Recognition of this data as Critical Energy Infrastructure Information (CEII)
- Robust protections for CEII
- Safeguards for commercially sensitive information such as:
 - o Pricing strategies
 - o Customer relationships
 - o Capacity plans and expansion timelines
 - o Technical specifications

Increased data-sharing requirements must be paired with equally strong data protections.

Stakeholder Process and Governance

To ensure operationally and economically viable compliance with the requirements of relevant authorities, NERC should:

- Require relevant authorities to engage directly with large load operators when developing standards
- Establish clear stakeholder participation processes that prioritize entities with direct operational relevance

III. Looking Ahead: Future (Technical) Standards

As NERC moves beyond initial “bridge” standards toward technical requirements, DPN emphasizes the importance of maintaining operational feasibility.

Voltage Ride-Through Standards

Future low-voltage ride-through (LVRT) requirements must be carefully designed to reflect current technological capabilities.

- Requirements that are too stringent could necessitate significant new technologies and impose substantial costs
- Poorly calibrated standards risk driving facilities out of certain markets

Ramp Rate Requirements

Ramp rate standards should account for real-world constraints, including:

- Supply chain limitations
- Permitting timelines

Overly rigid requirements could unintentionally limit deployment and flexibility

Preserve Operational Viability

Across all future standards should:

- Maintain flexibility for large loads to continue participating in demand response and curtailment programs
- Avoid requirements that undermine the economic or technical feasibility of operations
- Ensure standards are developed in partnership with industry
- Evaluate the impact of changing resource mixes and inverter-based resources on essential reliability services such as frequency response, voltage support, and system inertia

Conclusion

DPN supports NERC's efforts to improve coordination and visibility around large loads. However, the definition must only pertain to loads that have actual reliability impacts on the system rather than a targeted scope focused on industry rather than impact.

Beyond the definition, DPN recognizes that process-based standards are an appropriate first step.

However, success will depend on:

- Early and meaningful coordination with industry
- Streamlined and consistent data requirements
- Strong protections for sensitive information
- A continued focus on operational feasibility as standards evolve

A balanced approach will strengthen grid reliability while enabling the continued growth of critical digital infrastructure.



Proposed Changes to NERC Rules of Procedure “Computational Load Entity”

Comments of the Electricity Consumers Resource Council (ELCON)

The Electricity Consumers Resource Council (ELCON) respectfully submits these Comments in response to the National Electric Reliability Corporation’s (NERC) proposed revisions to the NERC Rules of Procedure posted April 1, 2026¹ incorporating defined terms and registration criteria for “Computational Load Entities.”

ELCON is the national association representing large industrial consumers of electricity. ELCON member companies create a wide range of products from virtually every segment of the industrial community—we own and operate hundreds of major facilities and are significant consumers of electricity throughout the United States. Reliable electricity supply at just and reasonable rates is essential to our members’ operations.

ELCON has been engaged in the work of the NERC Large Load Task Force (now Large Load Working Group or LLWG) since its inception in August 2024 and appreciates the efforts of NERC staff and industry to quickly anticipate and mitigate emerging risks to the Bulk Power System (BPS) posed by massive new loads interconnecting to the BPS at speeds our current system was not designed for.

Following the completion of two whitepapers,² NERC determined that “unforeseen and difficult-to-explain dynamic electrical behavior” made it appropriate “to propose a new functional entity type that focuses on loads with these characteristics [and] further define the entities that will be subject to registration based on their individual characteristics as well as the aggregate impact of multiple Computational Loads meeting the Registry

¹ <https://www.nerc.com/who-we-are/rules-of-procedure/proposed-changes-to-rules-of-procedure>.

² Characteristics and Risks of Emerging Large Loads (July 2025), <https://www.nerc.com/globalassets/who-we-are/standing-committees/rstc/whitepaper-characteristics-and-risks-of-emerging-large-loads.pdf> (Whitepaper #1); Assessment of Gaps in Existing Practices, Requirements, and Reliability Standards for Emerging Large Loads (March 2026), <https://www.nerc.com/globalassets/our-work/guidelines/reliability/white-paper---assessment-of-gaps.pdf> (Whitepaper #2).

Criteria.”³ NERC further correctly identified the need to incorporate certain minimum thresholds “to distinguish [] those entities, such as industrial manufacturers, with a ‘de minimus’ [sic] amount of Computational Load do not meet the criteria for registration as a Computational Load Entity. This distinction is based on these entities’ differing electrical behavior historically.”⁴

As explained below, ELCON is concerned that, despite NERC’s clear intent to register only those loads that exhibit certain behaviors that risk BPS stability and reliability, the proposed definitions in the Rules of Procedure and registration criteria in Section III of Appendix 5B (separately and jointly) could capture some traditional loads and, by requiring registration by end users, do not effectively target the entities responsible for ensuring reliable usage of the BPS.

In fact, nothing in the definition or registration criteria addresses the behaviors that introduce risk to the BPS, but rather focus on ancillary usage of a broad set of IT equipment that may or may not introduce those risks. Industrial and manufacturing large load customers are fundamentally different from the data center and computational facility activities that precipitated NERC’s large load work. In contrast, traditional manufacturing and industrial customers operate predictable, process-driven load profiles; are already subject to existing power quality expectations under IEEE 519;⁵ have decades-long interconnection histories with their host utilities; and do not exhibit the behaviors the LLWG identified as the primary drivers of BPS reliability risk. As proposed, the CLE definition and registration criteria do not adequately target those entities and behaviors identified in the whitepapers and the CLE Technical Reasoning.

Appendix 2 – Definitions Used in the Rules of Procedure

The proposed revisions to Appendix 2 of NERC’s Rules of Procedure introduce two new terms:

“Computational Load” means Load comprised of electric power demand from information technology equipment, such as servers, storage, and networking hardware.

“Computational Load Entity” means the end-user or the entity that hosts end-users that receives electric power for Computational Load.

As an initial matter, both proposed definitions are anchored exclusively to equipment type—“information technology equipment, such as servers, storage, and networking hardware”—rather than to any of the behavioral characteristics that NERC’s own LLWG analysis

³ CLE Technical Reasoning, p. 2 (Apr. 2, 2026), <https://www.nerc.com/globalassets/who-we-are/rules-of-procedure/proposed/cle-technical-reasoning-april-2026-posting.pdf>.

⁴ *Id.*, p. 3 (emphasis added).

⁵ IEEE Standard for Harmonic Control in Electric Power Systems.

identified as the source of BPS reliability risk.⁶ NERC’s technical work consistently grounds the large load reliability concern in specific operational behaviors: second-to-second demand oscillation tied to AI training workloads, voltage-sensitive customer-initiated load reductions (CILR), and synchronized cross-facility demand reductions. A definition anchored to equipment type rather than demonstrated or expected behavior cannot effectively distinguish load facilities that pose reliability risks from those that do not. ELCON urges NERC to anchor the definition to behavioral criteria, or at a minimum incorporate a substantial-proportion threshold as detailed below.

The “Computational Load Entity (CLE)” definition is independently problematic and overly broad. As worded, any user of electric power demand from information technology equipment satisfies this definition and is a candidate for registration. Taken to its extreme, anyone with a computer is a potential NERC registrant.

The phrase “the entity that hosts end-users” could capture a landlord or owner of a multi-tenant industrial park that has no direct control over individual tenants’ electrical behavior. Similarly, an industrial manufacturer that operates a vendor-managed server room could be registered as a CLE even though it has zero operational control over that equipment’s electrical behavior or demand variability. ELCON recommends that NERC revise the CLE definition to require that the entity have operational control over the computational load at the facility. An entity that passively receives electric power for use by a third-party operator of information technology equipment should not bear registration obligations arising from that operator’s equipment and behaviors.

In addition to the overly broad nature of the proposed definitions, it is unclear whether the definition and registration criteria are to be considered separately or jointly. In other words, must an entity first meet the explicit definition of a (CLE) before applying the registration criteria to determine if they are subject to registration? Is there a scenario where a “computational load entity” would not meet the MW and voltage registration criteria as defined in Appendix 5B of NERC’s Rules of Procedure?

For example, Company X manufactures widgets with a load profile of 28 MW and uses 1.6 MW of that demand to power its internal communications server (email), file database, internal management systems, process controls, and onsite data processing. Based on the definition alone, Company X could be considered a “Computational Load Entity” as some of its demand is used to power its information technology equipment.

However, if NERC’s intent is to directly tie the *characteristics* and *behaviors* of certain loads that pose a risk to the BPS and then apply the specific size and voltage thresholds to determine their registration eligibility, it is not entirely clear as written.

If NERC does not intend to capture traditional loads hosting informational technology that is ancillary to its primary industrial process, ELCON recommends that NERC modify the proposed definition of “Computational Load” to include the qualifying language “load comprised *primarily* of electric power demand from information technology equipment... .”

⁶ See Whitepaper #2.

Alternatively, NERC could include a percentage threshold to the definition. For example, “Computational Load means Load comprised of at least 50% of total electric power demand coming from information technology equipment... .”

With the clarification proposed by ELCON, traditional loads such as large commercial and industrial loads would not qualify for registration as the first criteria, the CLE definition, is not applicable. Thus, the proposed registration criteria in Appendix 5B would be irrelevant.

Appendix 5B – Statement of Compliance Registry Criteria

If, however, NERC intended to cast a broad net initially, as currently proposed in the CLE definition, with the registration criteria being more narrowly targeted to size impacts, the low thresholds in the proposed revisions to Appendix 5B of NERC’s Rules of Procedure would still inadvertently capture a wide swath of large loads that do not exhibit the characteristics and behaviors identified in the whitepapers and supporting documentation. Such broad implementation would be extremely inefficient and would distract industry from focusing on mitigating the behaviors that pose actual risk to BPS reliability.

Applying the proposed registration criteria in our Company X example, after meeting the broad criteria in the CLE definition of load with “electric power demand from information technology equipment,” Company X would then apply the registration criteria proposed in the revisions to Appendix 5B which specifies that a CLE must contribute at least **20 MW** in aggregate load at a single point of interconnection at a voltage of **60 kV** or higher where it hosts at least **1 MW** of load attributable to information technology equipment.

Here again, Company X (28 MW load, connects at 60 kV, where 1.6 MW of its power demand is used for internal information technology equipment) would meet each of the proposed criteria merely because of its size, not its operational characteristics. According to NERC’s Technical Reasoning supporting document, the 1 MW of computational load represents a de minimis amount that would distinguish traditional load “such as industrial manufacturers.”⁷ It is not clear how a facility with only 1 MW of computational load (defined only as “informational technology equipment”) would exhibit any of the variable load patterns attributed to reliability risks that precipitated NERC’s large load work. Even in aggregate where that 1 MW load was located with 19 other facilities with at least 1 MW of computational load (i.e. 20 MW in aggregate), there is no direct link that ties the risk of the behavior and the impact to the grid due to its size. This approach risks including traditional industrial, manufacturing, and commercial facilities—many of which utilize computational processes as an ancillary component of their operations—but whose load profiles, operating characteristics, and system impacts are well understood and already adequately addressed through existing planning, interconnection, and operational frameworks.

⁷ CLE Technical Reasoning, p. 3.

Critically, the proposal does not establish a clear, evidence-based nexus between the mere presence of “computational” activity and an increased or unique reliability risk to the BPS. Absent such a showing, expanding mandatory registration to these facilities would impose significant compliance obligations without a commensurate reliability benefit. NERC’s registration framework has historically been grounded in functional distinctions that align with demonstrated reliability impacts. Departing from that principle here could dilute the effectiveness of the registry by including entities that do not meaningfully affect system reliability.

Further, the 60 kV voltage threshold requires explicit legal and technical justification that NERC has not provided. FERC’s definition of the Bulk Electric System establishes 100 kV as the general bright line, with narrow exceptions justified by specific demonstrated reliability impacts. NERC’s prior extension to 60 kV for generator registration was grounded in documented technical analysis of sub-100 kV generation impacts on the BPS. NERC should be required to provide equivalent technical justification for applying load registration at 60 kV, and absent such justification, ELCON recommends the voltage threshold be set at 100 kV. Similarly, NERC should publish the reliability analysis, if any, that supports the 20 MW aggregate threshold, as the LLWG guideline itself explicitly states that no MW threshold is specified in that document. Registration thresholds must be grounded in demonstrated reliability impact, not arbitrary size proxies. Finally, the proposed aggregation approach—measuring load across a single point of interconnection—does not correspond to the actual reliability risk that NERC seeks to address. The LLWG’s documented concern is with cross-facility synchronized demand reductions across geographically dispersed facilities. A single 28 MW industrial facility with 1.6 MW of internal IT equipment presents no such cross-facility synchronization risk regardless of its size, and the proposed criteria fail to make this distinction.

Should NERC reject the proposed qualifying language in Appendix 2 and adopt the CLE definition as written, ELCON requests that NERC adjust the MW and voltage thresholds proposed in Appendix 5B to more closely align with actual risk impact to the BPS.⁸ Alternatively, the 1 MW “de minimis” criteria could be replaced with:

and 3) at least 50% of that aggregated connected Load capability is attributable to powering Computational Load.⁹

This revision also recognizes that NERC’s concern does not lie solely with the mere size of these loads and their variable behavior, but that the *proportion* of computational loads in relation to the entire system will significantly increase causing an even greater risk to the BPS.

⁸ The proposed 20MW is entirely too low and fails to target the size impacts of actual observed behavior, such as the often-cited incidents in Virginia and Texas that each involved loads of over 1GW.

⁹ ERCOT has recently recognized this distinction in its revised definition of “Large Computational Load” which includes the criteria that a 50% or greater portion of the demand at a site be attributable to electronic based load.

Finally, the breadth of the definition and registration criteria creates uncertainty for entities that may not reasonably have considered themselves subject to NERC jurisdiction, leading to potential over-reporting, administrative burden, and inefficient allocation of compliance resources—for both industry and NERC. Absent these refinements, the proposed criteria risk being both overinclusive and insufficiently tied to the reliability concerns it is intended to address. We respectfully urge NERC to revise the proposal to ensure that any expansion of the registration regime is targeted, justified, and proportionate to demonstrated risks to the BPS.

Registration of End-Users

NERC's jurisdiction over "users, owners and operators" was established by section 215 of the Federal Power Act as enacted through the Energy Policy Act of 2005. Since 2005, NERC has asserted that authority through its registration requirements, mandatory reliability standards, compliance and monitoring programs, and enforcement at the transmission level for those responsible for operating and planning the bulk power system. Although there has been some consideration of NERC's jurisdiction over certain distribution functions, there has never been an instance where NERC's authority over "users" has been challenged or more narrowly defined. Specifically, as "users" are typically under retail (state and local) jurisdiction, where does NERC's authority over "users" begin and end?

Size and voltage thresholds offer little protection when defining NERC jurisdiction as the end-user continues to evolve as does NERC's scope. The generally accepted bright-line for NERC jurisdiction has traditionally been 100 kV and above as memorialized in the Bulk Electric System definition in NERC's Rules of Procedure. With the introduction and increased penetration of variable resources, NERC determined that it needed to assert jurisdiction at the 60 kV in some instances¹⁰.

Here again, NERC is proposing jurisdiction at the 60 kV level without addressing the retail v. wholesale, bulk electric system, transmission v. distribution divide. Further, it is questionable to what extent an entity whose chief function falls well outside of electric grid service should be required to manage grid operations and protocols. The reliability concerns cited in the proposal—such as large, rapidly changing load profiles or novel operational characteristics—can be more appropriately and lawfully addressed through entities that are already subject to NERC oversight. Transmission Owners, Transmission Operators, Balancing Authorities, and Load-Serving Entities are already responsible for planning,

¹⁰ See NERC Rules of Procedure, Appendix 2 "'Generator Operator' means the entity that: 1) operates generating Facility(ies) and performs the functions of supplying energy and Interconnected Operations Services (Category 1 GOP); or 2) operates non-BES inverter based generating resources that either have or contribute to an aggregate nameplate capacity of greater than or equal to 20 MVA, connected through a system designed primarily for delivering such capacity to a common point of connection at a voltage greater than or equal to **60 kV** (Category 2 GOP)" (emphasis added).

interconnection oversight, operational coordination, and real-time balancing of load. These registered entities are well positioned to assess and manage the impacts of large or variable loads within their systems using existing or refined reliability standards, interconnection requirements, and operational tools. “Users” do not share the requisite expertise nor visibility and control into system operations. Their core mission is health care, manufacturing, chemical processing, mining, data processing, national security, logistics, etc. NERC must ensure that those entities with the technical expertise and responsibility for managing the grid are the entities responsible for ensuring responsible usage of the grid by their customers.

The existing NERC reliability framework places responsibility for BPS reliability squarely on registered entities—transmission owners, transmission operators, planning coordinators, balancing authorities, and distribution providers—who are professionally staffed, technically equipped, and legally obligated to manage BPS reliability. These entities already have the tools and authority to manage large load interconnections through FAC-001 interconnection requirements, qualified change triggers in interconnection agreements, and real-time operational coordination. The LLWG Reliability Guideline itself directs Transmission Owners to incorporate guideline recommendations into FAC-001 interconnection requirements; this mechanism can achieve the same reliability outcomes without the jurisdictional overreach of direct end-user registration. ELCON urges NERC to strengthen the obligations of existing registered entities rather than conscript retail customers into the NERC registry for purposes of addressing load behaviors that existing registered entities are already positioned and obligated to manage.

Reliability is essential to all of us and time is critical as our Nation’s energy system evolves rapidly on both the user and producer side. To swiftly address the risks and regulatory gaps identified by the LLWG, the recommendations in the Level 2¹¹ and Level 3¹² alerts should be formalized through revisions to existing standards (bridge standard) with development of new standards to be applicable to current registered entities.

ELCON appreciates the extensive outreach and analysis devoted to ensuring our grid can meet the challenges of integrating emerging large loads and large load technologies. ELCON looks forward to continuing our engagement with NERC as we address large load operations and the criticality of reliable electricity.

¹¹ Industry Recommendation: Large Load Interconnection, Study, Commissioning, and Operations (Sept. 9, 2025),

¹² Essential Action to Industry: Computational Load Modeling Studies, Instrumentation, Commissioning, Operations, Protection, and Control (May 4, 2026), <https://www.nerc.com/globalassets/programs/bpsa/alerts/level-3-computational-load-alert.pdf>.

/s/ Karen Onaran
Karen Onaran
President & CEO
Electricity Consumers Resource Council (ELCON)

May 15, 2026

Good afternoon,

For the proposed term, “Computational Load Entity”, the new entities should be designated as either a “Computational Load Owner” or “Computational Load Operator” like the existing functions of TO & TOP and GO & GOP. One organization can be registered as both, or multiple organization working on a project can register separately based on their scoped roles. This allows the appropriate reliability standard requirements to be assigned to an entity that owns and/or develops the project and to the entity that will be operating the Computation Load Facility.

Thank you,
Cassie Sims
Sr. Lead Analyst, Security & Reliability Programs

entergy.com



NERC Rules of Procedure Team –

Please see below for the comments from AECC's SMEs regarding the proposed revisions to the Rules of Procedure:

Computational Load Definition:

- The Standards Drafting Team should consider revising the definition of Computational Load to ensure it clearly reflects the explicit exclusion of equipment that is used by direct physical/local interaction (e.g., laptop computers).
- The Standards Drafting Team should consider revising the definition of Computational Load to ensure it is clearly restricted to server type hardware only (which would include "workstation" type hardware that is operated in a server mode or arrangement) whose power usage from the transmission system can be directly or indirectly controlled. This control can be due to the inherent processing source for the equipment between the transmission system and some other on-site source (e.g., generator, UPS, etc.). The ability to control the load seen by the transmission system will likely be an important component in future NERC Reliability Standards. If the owner cannot control the equipment, NERC will not be able to mandate any requirements that seek to control it.
- To summarize: Computational Loads should be non-portable, fixed, server type information technology equipment whose transmission system power usage can be controlled.

Other Thoughts:

- The Standards Drafting Team should reconsider the 1 MW threshold. 1 MW of load seems too low, and it may pull smaller data centers which do not pose a problem for the transmission system into scope. NERC should explain its reasoning and provide a justification for this specific threshold. The criteria should focus on the amount of computational load only, and not the amount of other traditional loads.
- Is NERC's intention to focus on the type of equipment or the type of load, and not the characteristics of the load? Are the concerns based more upon the fluctuations and fast ramp rates of loads regardless of load type? NERC should consider creating a "volatile loads" definition that is defined by the speed and degree to which the load magnitude changes instead of creating a "computational load" definition.

Thanks,

Ayslynn McAvoy

NERC Compliance Manager
Arkansas Electric Cooperative Corporation
Arkansas Electric Cooperatives, Inc.

To whom it may concern,

FirstEnergy will be supporting the comments generated by the Trade Association. They are attached to this email. We appreciate NERC taking our position into consideration.

Thank you,
Sarah Miller



Sarah Miller
Asst. Compliance Specialist

Hello NERC team,

Google agrees with the comments submitted by the Data Center Coalition (DCC) attached to this email.

Getting the definition details right is critical at this stage. The DCC has flagged several critical clarifications that will help NERC and industry ensure this effort is well targeted for the perceived risks.

Mike Della Penna | Global Infrastructure & Energy |

Dear NERC Team,

Thank you for the opportunity to provide stakeholder input on the Proposed Revisions to the Rules of Procedure and for NERC's continued efforts to address reliability considerations associated with emerging large/computational loads.

As outlined in the attached OEB–NERC MOU and IESO–NERC–NPCC MOU, these agreements define the framework for administering NERC reliability standards in Ontario, including clear roles and compliance accountabilities. Under this framework, the IESO is solely accountable to NERC, while Ontario market participants/entities comply through the IESO in accordance with the Market Rules rather than through direct registration with NERC. Given that changes to the Rules of Procedure may require direct registration in the NERC Compliance Registry, it is important that any revisions preserve this established compliance structure.

The following are our comments:

- Any revisions to NERC Registration Criteria to require registration of Computational Load Entities with NERC should respect existing jurisdictional registration frameworks for non-US entities.
- Clarification is needed regarding the meaning and intended application of “single point of interconnection,” since a Computational Load Entity that receives multiple supplies from the BPS may not meet the definition.
- Further clarification would be helpful regarding registration of multi-tenant and colocation facilities where the host facility and end-use tenants may have different levels of control over IT load behavior, such as a data centre owner operating the electrical infrastructure while tenants control server dispatch or workload migration.
- It would be helpful to understand the technical and/or reliability rationale for the proposed registration thresholds, including the 20 MW aggregate connected load threshold and the 1 MW Computational Load criterion, and whether these values are based on BPS impact analysis, operational experience, historical disturbance data, alignment with existing registration thresholds, or specific reliability risks intended to be screened.
- Clarification would be helpful regarding applicability to distribution-connected computational loads, including cases where the IT load is connected below 60 kV but the distribution system is supplied from transmission at or above 60 kV; consideration

could be given to whether applicability should be based on the immediate customer connection point, the upstream transmission connection point, or demonstrated BPS impact.

- Additional clarity would be helpful on the treatment of behind-the-meter generation, self-supply, and export-capable facilities for registration purposes. Depending on whether registration is based on gross load, net load, maximum connected load capability, or another metric, registration criteria may or may not be met.
- Clearly defining registration boundaries between CLEs and existing registered entities would help avoid uncertainty, duplication, or conflicting obligations, including interactions with Transmission Planners, Transmission Operators, Distribution Providers, Planning Coordinators, and Load-Serving Entities.
- Looking beyond the current effort to revise the ROP, additional clarity would be helpful on registration timelines, transition periods, and expectations for existing versus new computational load facilities.

In closing, we thank NERC for its continued attention to these important matters. We welcome further discussion should any points require clarification or additional consideration.

Sincerely,

Independent Electricity System Operator (IESO)
Compliance Assurance

Pirouz Honarmand | Specialist, Compliance Assurance | Legal Resources & Corporate Governance
Independent Electricity System Operator (IESO) |



One South Wacker Drive | Suite 1500 | Chicago, Illinois 60606

May 15, 2026

Invenergy Comments on Proposed Computational Load Entity RoP Revisions

NERC,

Thank you for the opportunity to provide comments on this important matter. Invenergy believes the proposed registration criteria for Computational Load Entities is overly broad, and the low MW thresholds may unintentionally require industrial manufacturers or other non-data center loads with a de minimis amount of Computational Load to register with NERC. This will place an unnecessary burden on both the newly registered entities and the Regional Entities as critical resources are dedicated to smaller facilities that do not present the same risk to the Bulk Power System as larger data centers.

Invenergy recommends NERC revise the proposed registry criteria to better align with the existing, and in development, thresholds established by RTOs and ISOs. Their criteria better reflect the load capabilities that they consider impactful to their systems and will ensure that resources are not spread thin. If NERC is determined to include loads of 20 MW or greater, we recommend a ratio-based threshold for the Computational Load capability that limits registration to those entities whose primary purpose is Computational Load.

Sincerely,
Rhonda Jones
Vice President of Reliability Compliance, Invenergy
Member, Reliability and Security Technical Committee

ISO/RTO Council Standard Review Committee's Comments on Proposed Computational Load Entity ROP Revisions

May 15, 2026

The ISO/RTO Council (IRC) Standards Review Committee¹ (SRC) offers the following comments on the proposed revisions to the NERC Rules of Procedure (ROP) to define Computational Load and register Computational Load Entities (CLEs), released on April 1, 2026. The SRC appreciates this opportunity to provide feedback on the proposed ROP revisions.

Summary of SRC Comments

- Consider developing additional definitions for Bulk Electric System-level Computational Loads, analogous to Category 1 Generator Owners and Generator Operators.
- Increase the 1 MW Computational Load threshold in the CLE registration criteria or replace it with a threshold based on what percentage of the CLE's total Load is Computational Load.
- Align the Distribution Provider (DP) registration threshold with the CLE registration threshold.
- Evaluate the best way to ensure NERC-level obligations apply to the interconnection of a CLE that pushes a distribution entity above the DP registration threshold.
- Revise the Computational Load definition to read as follows: *"Computational Load" means Load that uses electric power for equipment needed for operation of computers or computer programs. Examples of such equipment include, but are not limited to, processors, memory, data storage devices, and networking equipment.*
- Revise the CLE definition as follows: *"Computational Load Entity" means the ~~end-user or the entity-operator of a facility~~ that ~~hosts end-users that receives~~ uses electric power for Computational Load.*

SRC Comments

CLE Registration Threshold

The ROP revisions propose to require a CLE to register with NERC if the CLE contributes to an aggregate connected Load capability greater than or equal to 20 MW at a single 60 kV or greater point of interconnection to the Bulk-Power System and hosts 1 MW or more of Computational Load. The SRC offers no comment on the MW value used in the proposed 20 MW threshold, as a variety of thresholds exist in RTO/ISO-specific definitions of what constitutes a large load or CLE-type entity.

¹ For purposes of these comments, the SRC comprises the California Independent System Operator Corporation (California ISO), Electric Reliability Council of Texas, Inc. (ERCOT), ISO New England Inc. (ISO-NE), Midcontinent Independent System Operator, Inc., (MISO), New York Independent System Operator, Inc. (NYISO), PJM Interconnection, L.L.C. (PJM), and Southwest Power Pool, Inc. (SPP).

The SRC notes the proposed threshold appears to align with the threshold used for Category 2 Generator Owners and Generator Operators. Consequently, NERC may wish to consider whether it would be prudent to also establish a higher set of thresholds for CLEs that might be considered part of the Bulk Electric System, analogous to Category 1 Generator Owners and Generator Operators.

However, the SRC believes the 1 MW Computational Load threshold as proposed is too low and could result in registration requirements being imposed on entities that do not primarily perform Computational Load-type activities and cannot reasonably be considered Computational Loads (for example, a 200 MW manufacturing facility that happens to have 1.5 MW of on-site computing equipment). Establishing a higher threshold or a threshold based on a percentage of the total Load (e.g., more than 60% of the Load at the facility is Computational Load) would help prevent the definition from being overinclusive.

DP Registration Threshold

While the proposed ROP revisions do not presently address the DP registration threshold, the SRC recommends revising the DP threshold as part of this project. The original DP threshold was 25 MW, and the threshold was increased to 75 MW in 2015 to reduce the regulatory burden on smaller, lower-risk distribution systems. However, as the Standards Drafting Team in Project 2026-02 has recognized, DPs have an important role to play in reliably interconnecting CLEs to the Bulk-Power System. To avoid a scenario in which a CLE attempts to interconnect with an entity that is not registered as a DP and therefore has no NERC-level obligation to fulfill the DP's role in reliably interconnecting CLEs (for example, a 25 MW CLE seeking to interconnect with an entity that operates a 40 MW distribution system, resulting in a total system size of 65 MW, which is still below the current 75 MW DP threshold), the DP registration threshold should be revised to match the CLE registration threshold, regardless of whether that threshold is ultimately 20 MW or some other value.

Regardless of which registration thresholds are ultimately used for CLEs and DPs, NERC and the Project 2026-02 Standard Drafting Team will need to carefully consider the most effective way to address a scenario in which a CLE interconnection pushes a distribution system above the DP registration threshold (for example, under current thresholds, a 50 MW CLE seeking to interconnect with a 50 MW distribution system, resulting in a total system size of 100 MW). System reliability would be best served if the distribution entity were registered as a DP to fulfill the relevant NERC-level DP obligations before interconnecting the CLE. Achieving this result would likely require additional revisions to the DP registration criteria.

Computational Load Definition

The proposed Computational Load definition seem underinclusive and vulnerable to arguments about whether the examples provided (servers, storage, and networking hardware) limit the scope of the definition, thereby potentially excluding hardware used for advanced computing (such as computational accelerators or quantum computing hardware) or hardware that does not serve a traditional server-type role, such as hardware used in a peer-to-peer architecture instead of a server-client architecture. To better futureproof the definition and clarify the scope and role of any examples provided, the SRC recommends the Computational Load definition to be revised to read as follows:

“Computational Load” means Load that uses electric power for equipment needed for operation of computers or computer programs. Examples of such equipment include, but are not limited to, processors, memory, data storage devices, and networking equipment.

CLE Definition²

In the proposed CLE definition, the phrase “entity that hosts end-users” is somewhat vague, and the phrase “end-user or the entity that hosts end-users” creates ambiguity as to whether the end user or the facility host would be the CLE for a given facility, which in turn creates ambiguity about who is required to register if the facility meets the applicable CLE registration thresholds. It would be less ambiguous to assign registration responsibility to the facility operator in all cases, since the operator is best situated to ensure the facility complies with all applicable Reliability Standards. Consequently, the SRC recommends revising the definition as follows:

“Computational Load Entity” means the ~~end-user or the entity~~ operator of a facility that ~~hosts end-users that receives~~ uses electric power for Computational Load.

Conclusion

The SRC appreciates the opportunity to comment on the proposed ROP revisions. The SRC would welcome the opportunity to further discuss these comments or other Computational Load-related topics with NERC as appropriate and looks forward to engaging in the remainder of the ROP revision process.

² PJM is also submitting separate comments regarding the CLE definition and associated registration criteria.

Microsoft Comments on Proposed Revisions to NERC ROP: Computational Loads

5/15/2026

INTRODUCTION

Microsoft operates more than 400 data centers across North America and five continents. A reliable grid is essential to our ability to serve customers and support United States leadership in artificial intelligence. We welcome NERC's efforts to address the reliability risks posed by large computational loads and submit these comments to ensure the proposed framework is well-calibrated to that goal.

These comments build on Microsoft's prior submission on NERC's Standard Authorization Request for Project 2026-02, filed April 30, 2026, and focus specifically on whether the proposed definitions and registration criteria appropriately reflect system impact and reliability considerations.

COMMENTS

A. Existing Registered Entities Can Provide the Quickest Path to Mitigating Identified Risks

The existing reliability framework already provides effective tools for managing large load risks. Transmission Owners and Operators hold federal and state tariff authority that can be used to impose requirements on large load customers directly, i.e., no new registration category is required to close any accountability gap. Expanding the obligations of existing registered entities is faster to implement, avoids the complexity of bringing an entirely new class of entities into NERC compliance, and builds on processes that are already operational.

To make that approach effective, however, registered entities need to do their part. Microsoft recommends the drafting team establish clear expectations for existing entities to coordinate with large loads on telemetry standards, modeling data requirements, and grid-side monitoring—so that any compliance obligations placed on large loads are grounded in information that is available and actionable.

B. NERC Should Include All Large Loads That Impact the Bulk Power System, Not Only Computational Loads

Singling out "computational" load risks creating a framework that addresses one category of risk while leaving comparable risks from other large load types unaddressed. Large electrified industrial processes, battery manufacturing facilities, and similar loads can present the same reliability challenges—high ramp rates, concentrated interconnection, potential for correlated loss events—without falling within the proposed definition. A standard grounded in behavior and system impact rather than end-use would be more durable and more effective. At a minimum, NERC should commit to addressing other large load types in Phase 2.

C. The Definitions of "Computational Load" and "Computational Load Entity" Should Be Revised to Distinguish Between Differing Facility Operations and Load Behaviors

The definition of "Computational Load" as power demand from "information technology equipment, such as servers, storage, and networking hardware" is broad enough to capture hospitals,

universities, industrial facilities, and commercial buildings that use IT equipment as part of ordinary operations. These are stable, predictable loads that bear no resemblance to the reliability risks NERC has documented. The open-ended “such as” language compounds the problem by leaving the scope of the definition unclear for both registrants and auditors. NERC should ground the definition in the operational characteristics—load variability, ramp rate, power factor behavior—that actually drive reliability concerns and should include a clear exclusion for traditional loads that happen to use similar equipment.

The “Computational Load Entity” definition also does not reflect how data centers are actually structured. In many deployments, the entity that owns and finances the facility is entirely separate from the tenant that installs and operates the computational equipment, and it is the tenant’s operational decisions that drive the load behavior of reliability concern. The single “CLE” designation obscures this distinction and risks misallocating compliance obligations between parties that have very different relationships to the grid.

Microsoft recommends that NERC establish separate Computational Load Owner (“CLO”) and Computational Load Operator (“CLOP”) functional entities, modeled on the existing Generator Owner/Operator and Transmission Owner/Operator frameworks.

For example, the CLO, the entity that owns and maintains the facility and its interconnection infrastructure, would bear responsibility for facility-level obligations: design specifications, interconnection compliance, and equipment standards. The CLOP: the entity that controls real-time load behavior, would bear responsibility for operational obligations: load ramp management, communications with the Transmission Operator and Balancing Authority. A single entity can hold both registrations where it owns and operates the same facility. Where ownership and operation are split, as is common in co-location and wholesale lease arrangements, each party’s obligations would be delineated by function. This structure does not eliminate compliance responsibility for large load customers, but it ensures that each requirement attaches to the entity actually in a position to fulfill it, and creates a commercially workable framework for allocating grid compliance responsibilities between landlords and tenants.

D. The Aggregate Capacity and Load Thresholds Are Too Low and Risk Increasing Administrative Burden for Little Corresponding Reliability Benefit

The proposed 20 MW aggregate threshold captures a very wide range of facilities, many of which will have no meaningful impact on BPS reliability. NERC has not provided a technical basis for why a load at that level presents material risk, and the documented reliability incidents—involving load rejection events of 1,000 MW or more—do not support a 20 MW floor. The 1 MW computational load sub-threshold makes this worse: a facility with 1 MW of servers within a 20 MW footprint simply does not exhibit the characteristics NERC is trying to regulate. It is also important to note that what is observed at the point of interconnection is not the same as what is measured at the facility level. A load that appears behaviorally significant at the facility may present as routine variation at the POI on a larger system.

Microsoft recommends that NERC either raise the thresholds to levels supported by technical analysis and demonstrated material impact on the Bulk Power System or simplify to a single Computational Load threshold that directly reflects the characteristic driving reliability risk to the Bulk Power System. Registration thresholds should reflect what is observable and meaningful at the Bulk Power System level, not just the size of the facility.

E. Any New Category of Registered Entity Should Include Explicit Exclusions for Existing Facilities and Facilities in Advanced Design

Facilities that are already in service, under construction, or in advanced development were designed and financed without any expectation of NERC registration or technical requirements, which may have meaningful impacts on sites that are operational or in advanced design. Applying new requirements to them retroactively risks stranded investment and project disruption. Microsoft recommends that NERC includes a clear mechanism for requesting an exemption from reliability standards or clear grandfathering provisions covering facilities currently in service; facilities under active construction; facilities in design, permitting, or procurement; and facilities with executed interconnection or service agreements. Where new requirements do apply, compliance timelines should be developed in consultation with affected entities before they are finalized.

Microsoft recognizes that certain existing large computational loads may present genuine near-term reliability risks that NERC has a legitimate interest in addressing promptly. We do not categorically dispute that proposition. Our concern is with the remedy, not the risk assessment. The proposed framework applies registration and compliance obligations broadly, without distinguishing between facilities that present material BPS risk and those that do not. A 500 MW hyperscale AI training cluster interconnected directly to the transmission system presents a fundamentally different risk profile than a 25 MW colocation facility serving mixed commercial tenants. Applying the same retroactive compliance mandates to both, on the same timeline—is not proportionate to the actual reliability exposure involved.

The primary position is that any new registration requirements should apply prospectively, with existing and in-development facilities protected by clear grandfathering provisions. Facilities currently in service, under active construction, in design or procurement, or with executed interconnection or service agreements were designed and financed without any expectation of NERC registration; retroactive application risks significant stranded investment or substantive additional investment to bring facility into compliance.

In the alternative, if NERC determines that some existing facilities must be brought into compliance, Microsoft urges that the compliance pathway be tightly calibrated to the risk each facility actually poses to the Bulk Power System. Obligations and timelines should reflect the facility's size, interconnection voltage, load variability characteristics, and proximity to grid constraints - not a uniform mandate applied to every facility above a low MW threshold. This approach is consistent with the definitional precision we advocate in Sections B, C, and D: a framework that is appropriately scoped to the problem NERC has identified will be more durable, more defensible, and more likely to achieve its reliability objectives.

F. The “Point of Interconnection at ≥ 60 kV” Criterion Requires Clarification to Prevent Inconsistent Application

The proposed criterion—interconnection “at a single point to the Bulk Power System at a voltage ≥ 60 kV”—is ambiguous in common facility configurations where a step-down transformer sits between the load and the transmission system. A load entity could plausibly argue it interconnects at the lower secondary voltage and falls outside the registration threshold, even where the functional BPS connection is clear. NERC should close this gap by adopting language modeled on the existing Category 2 inverter-based resource definition, which captures facilities “connected through

a system designed primarily for delivering capacity to a common point of connection at a voltage ≥ 100 kV.” That language is already accepted by industry and regulators and would apply straightforwardly here. A 100 kV threshold also aligns with NERC’s existing Bulk Electric System definition framework and provides greater consistency across reliability registrations.

CONCLUSION

Microsoft appreciates NERC’s consideration of our comments and a shared focus on addressing reliability issues in an expedited manner. Microsoft’s recommendations are designed to encourage a framework that is based on material risk, existing regulatory structures, and technical rigor.

In summary, Microsoft respectfully recommends NERC:

- Maximize the use of existing Registered Entities to mitigate large load reliability risks, ensuring those entities provide the data, telemetry, and technical guidance that large load compliance will require;
- Expand the scope of the ROP revisions to address all large loads with material BPS impact, or commit to doing so in Phase 2;
- Revise the definitions of “Computational Load” and “Computational Load Entity” to reflect operational behavior and system impact rather than equipment type, and adopt a Computational Load Owner/Operator framework consistent with established NERC functional entity structures;
- Recalibrate the aggregate MW and Computational Load sub-thresholds upward, supported by clear technical analysis demonstrating material BPS impact at the selected levels;
- Include explicit grandfathering provisions covering in-service, in-construction, and in-design facilities, as well as those with executed agreements, with realistic compliance timelines developed in consultation with affected entities; and
- Clarify the “point of interconnection at ≥ 60 kV” criterion using language modeled on the existing Category 2 IBR definition to prevent inconsistent application and registration avoidance, while aligning the threshold with NERC’s existing BES framework.

To who it may concern,

The proposed computational load definition would benefit from additional clarification in two areas. First, the reference to “storage” is unclear and should specify whether it refers to energy storage resources, data storage infrastructure (e.g., servers or hard drives), or another type of storage entirely. Second, additional clarity is needed regarding the phrase “aggregate connected Load capability” in relation to “a single point of interconnection” in Appendix 5B, page 8. Specifically, it is unclear whether the intent is for the aggregated connected load to exist behind a single POI in order to meet the registration threshold. For example, if multiple adjacent POIs each serve approximately 19 MW of load, it is uncertain whether those loads would collectively be considered for Compliance Registry applicability or treated independently below the threshold

Adrian Harris

Advisor / Regulatory Compliance

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NYPA supports the comments submitted by the Trade Associations (APPA, LPPC, EEI, NRECA and TAPS) on the proposed revisions to NERC Rules of Procedure.

Libin Varghese

Engineer, Reliability Standards & Compliance

New York Power Authority

123 Main Street, 11th Floor (11-A)

White Plains, NY 10601

www.nypa.gov

To whom it may concern:

I am emailing to express Oracle's support for the comments being submitted today by the Data Center Coalition on the proposed Rules of Procedure changes with definitions and registration criteria for Computational Loads, attached for reference. Oracle is a Registered Ballot Body participant within Segment 7.

Sincerely,
Madeline Fleisher

The Oracle logo, consisting of the word "ORACLE" in a bold, red, sans-serif font.

Madeline Fleisher | Senior Managing Counsel
Oracle Legal



May 15, 2026

RE: PJM's response to NERC's Rule of Procedures (ROP) revisions related to Computational Loads.

Hello,

Background to the following comments: In PJM's experience, many of the entities that build and own data centers are essentially landlords, who lease out the data center facility to an anchor tenant or multiple tenants. As a result, the entity that owns the physical facility and interfaces/contracts with the distribution utility may be separate and distinct from the entity that controls the computational equipment therein. The draft ROP changes to Appendix 5B appear to capture this concept better than the definition in the currently drafted Computational Load Standard Authorization Request¹, by defining Computational Load Entity as “[t]he end-user or the entity that hosts end-users that receives electric power for Computational Load.” Whichever definition is ultimately adopted, the goal should be the same—to provide the Regional Entity with clear authority to ensure that contractual relationships between the computational load and the host do not frustrate the Regional Entity's ability to register the party who presents the actual risk to the BES.

PJM notes that the draft revisions to Appendix 5B establish that a Computational Load Entity will be registered only if it: 1) contributes to an aggregate connected Load capability greater than or equal to 20 MW; 2) at a single point of interconnection to the Bulk Power System at a voltage greater than or equal to 60 kV; and 3) hosts 1 MW or greater of Computational Load. PJM recommends increasing the load threshold from 20 MW to 50 MW. This is the threshold by which PJM defines a large load and will “right size” any subsequent compliance obligations to avoid capturing a wide range of entities that do not present the same risk to BES reliability as larger data centers. Increasing this threshold will also permit Regional Entities to focus their resources on auditing only those Computational Load Entities that present the most risk to the BES, in line with the overall policy trajectory of the risk-based Reliability Assurance Initiative. PJM correspondingly recommends reducing the DP threshold to greater than 50 MW of peak Load that is directly connected to less than or equal to 60kV. The original DP threshold was 25 MW, and the threshold was increased to 75 MW in 2015 with the aim of reducing the regulatory burden on smaller, lower-risk distribution systems. While this may have made sense in 2015, PJM is now dealing in real time with the challenges of the increased DP threshold, such as collecting IBR and DER data now required by MOD Standards (and soon to be TOP and IRO

¹ Project 2026-02 Computational Loads Standard Authorization Request:
https://www.nerc.com/globalassets/standards/projects/2026-02/sar-comment-period/2026-02-computational-load-alignment-phase-1-sar_sc-approved.pdf



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Audubon, PA 19403

Standards) where the host DP is not a NERC registered entity. Aligning the Computational Load Entity threshold with the DP registration threshold will ensure a level playing field from a compliance and implementation perspective. In addition, NERC must be wary of a scenario in which multiple data centers exist at the same campus, but they are physically segregated to fall below 50 MW each, so that none of them are required to register as a Computational Load Entity. To address this, NERC should consider the “one-mile” rule that FERC has used for many years regarding Qualifying Facilities under PURPA. Under this rule, a one-mile radius is drawn around a given facility to ensure that entities cannot escape regulatory requirements by breaking themselves into smaller parts. See 18 C.F.R. § 292.204.

Please feel free to contact me should you have questions regarding this matter.

Sincerely,

Elizabeth (Becky) Davis
Sr. Lead Analyst, FERC & Reliability Compliance

cc: Thomas Devita

To Whom it may concern,

Eversource would like to voice its support for the joint comments filed by the American Public Power Association (APPA), Edison Electric Institute (EEl), Large Public Power Council (LPPC), National Rural Electric Cooperative Association (NRECA), the Transmission Access Policy Study Group (TAPS), and the National Association of Regulatory Utility Commissioners (NARUC).

Eversource supports the comments submitted by EEl and the trades, with one exception: we oppose limiting the registration of Computational Loads (CL) exclusively to those “75 MW connected at $\geq 100\text{kV}$ for Computational Load Facilities, consistent with the NERC Bulk Electric System definition for dispersed power producing resources (i.e., inverter-based resources) – where the current criteria of greater than 75 MVA at a common point of connection at $\geq 100\text{kV}$ is utilized”. Instead, Eversource aligns with the Trades’ alternative proposal for a model similar to Generator Owner/Operator, establishing Category 1 and Category 2 classifications. Eversource and its Regional Transmission Organization currently plan to include all large loads ($>20\text{ MW}$) in regional planning and study processes. Adopting this two-category approach allows for a risk-based compliance structure, ensuring uniform reliability requirements for all large loads, while allowing stricter reliability and security standards for Category 1 when warranted and technically justified.

Thank you and please reach out if you have any questions or comments.

Respectfully,

Joshua London
Senior Engineer, Reliability Compliance

EVERSOURCE

Good morning,

Consumers Energy is submitting comment on the Proposed Computational Load Entity ROP Revisions.

Neither NERC's *Assessment of Gaps in Existing Practices, Requirements, and Reliability Standards for Emerging Large Load* nor the findings in the *NERC Aggregated Report on NERC Level 2 Industry Recommendation: Large Load Interconnection, Study, Commissioning, and Operations* provide sufficient justification for establishing the proposed registration thresholds of an aggregate load of 20 MW with a 1 MW Computational Load component.

While NERC's publications to date have demonstrated the potential reliability risks associated with large loads—particularly Computational Loads—Consumers Energy agrees that the creation of a new registered entity is appropriate. However, Consumers Energy does not agree with the proposed thresholds. Rather than defining both an aggregate load threshold and a Computational Load sub-threshold, NERC should simplify the criteria by establishing a single threshold based on total Computational Load, as this load characteristic has been identified by NERC as posing the greatest potential risk to the Bulk Power System (BPS).

The Level 2 Alert report indicates that approximately 90–95 percent of existing industry-defined large load thresholds are at or above 20 MW. However, the report does not demonstrate that a load at or near 20 MW, in and of itself or in reasonable local aggregation with similar loads, presents a material risk to BPS reliability. NERC recently revised their Rules or Procedure to include IBR generators 20 MW or larger interconnecting at 60 kV or higher and demonstrated that, in aggregate, these units could impact the BPS. Therefore, it is logical that NERC would propose a similar threshold for Computational Load Entities, however, NERC must acknowledge the fundamental difference in quantity and magnitude of interconnections between dispersed IBR generation resources expected to interconnect to the system and Computational Loads expected to interconnect to the system. Consumers Energy recommends that NERC consider increasing the proposed threshold to better align with thresholds already established—or currently being proposed—by Regional Transmission Organizations.

As currently defined, Consumers Energy does not believe the proposed criteria sufficiently

distinguish between traditional industrial customers with de minimis Computational Load and entities whose primary business purpose is Computational Load. While the proposed 1 MW Computational Load threshold may have been derived from responses to NERC's Level 2 Alert questionnaire, this value is unreasonably low if the intent is to capture entities whose Computational Loads are expected to have a meaningful impact on the BPS.

If the Computational Load threshold is increased to a more appropriate level, Consumers Energy recommends removing the proposed 60 kV voltage threshold for clarity. The reliability gaps identified by NERC are not inherently tied to a specific interconnection voltage level, but rather to the size and characteristics of the aggregated load. If NERC determines that retaining a voltage threshold is necessary, Consumers Energy recommends including clarifying language such as: *"at a single point of interconnection directly to the Bulk Power System, or through facilities primarily designed to deliver power to the Computational Load Entity from a common point of connection, at a voltage greater than or equal to XX kV."* This clarification would mitigate potential interpretation issues. For example, scenarios exist where the Transmission Owner owns a networked transmission system, the Distribution Provider interconnects one or more 138/34.5 kV transformers, and the Computational Load Entity could assert that its facilities interconnect at 34.5 kV and are therefore exempt from registration. Incorporating language similar to NERC's BES Inclusion I4 would help prevent such gaps in interpretation.

Regards,
Eric R. Sutlief

Eric R. Sutlief, FERC/NERC Regulatory Compliance



Hello,

After discussing with our neighboring utilities, PacifiCorp would like to submit the following minor proposed modifications to item III.c.1 in Appendix 5B:

III(c) Computational Load Entity: III.c.1 Computational Load Entity that 1) contributes to an aggregate connected Load capability greater than or equal to 20 MW; 2) ~~aggregates to at~~ a single point of interconnection ~~or substation on to~~ the Bulk Power System ~~operated~~ at a voltage greater than or equal to 60 kV; and 3) hosts 1 MW or greater of Computational Load.

Many thanks,
Lindsay

Lindsay Wickizer

NERC Compliance

PacifiCorp

825 NE Multnomah, Suite 2000

Portland, Oregon 97232

Good morning,

Southern Company offers the following comments for consideration:

“Southern Company supports the comments from the joint Trade Associations (APPA, EEI, LPPC, NRECA, and TAPS) on the proposed Rules of Procedure modifications pertaining to Computational Loads.”

Best regards,

Laura Winston
NERC Compliance Coordinator
Operations Compliance
Southern Company Services

Comments on Proposed Changes to NERC Rules of Procedures Regarding Computational Loads

The Solar Energy Industries Association (SEIA) values the opportunity to provide comments on proposed changes to NERC Rules of Procedures (ROP) regarding Computational Loads.

SEIA is a trade organization of over 1,200 members in the solar PV and storage industry. Many of our members are involved in creating solutions that provide electric power to Computational Loads. This includes solar PV and Battery Energy Storage Systems (BESS) which operate through Power Purchase Agreements (PPAs), market participation, and co-location with Computational Loads. It also includes the stand-alone supply of BESS co-located with Computational Load, which can provide multiple value streams to Computational Load. In addition, some members are evaluating if resources co-located with Computational Load can provide services to the grid.

SEIA and its members are committed to working with NERC to maintain Bulk Power System (BPS) reliability.

We offer the following comments, arranged by ROP section.

Appendix 2

Proposed Terminology “Computational Loads”

As proposed, “Computational Load means Load comprised of electric power demand from information technology equipment, such as servers, storage, and networking hardware.” We agree with NERC staff that a term other than “Large Load,” without including additional descriptors for the electronic loads like data centers, AI centers, and crypto-currency mining, is not a satisfactory term. “Large Load” also applies to large process facilities, oil and gas loads, education and medical campuses, pharmaceutical loads, and manufacturing loads. These large loads do not contain the same amount of servers, data storage, and networking equipment, nor the same operational characteristics as the loads which are the subject of NERC’s “Large Load Action Plan.”

However, the term “Computational Load” to describe loads such as data centers, AI training and inference facilities, and cryptocurrency mining facilities may not be sufficient. The proposed “Computational Load” definition includes “Load comprised of electric power demand from information technology equipment.” However, most Bulk Electric System (BES) loads contain a Computational Load component and would therefore be included in the definition, which is not the intention of the “Large Load Action Plan”.

The definition also contains the term “**such as** servers, storage, and networking hardware.” The term “such as” is non-inclusive and introduces ambiguity as to the equipment that may, or may not be, included in the definition.

In addition, the term “storage” should be amended to “data storage”, to clarify that it is referring to information technology equipment such as Network Attached Storage (NAS), Hard Disc Drives (HDDs), and Solid State Devices (SSDs) and not “energy storage” such as BESS.

Furthermore, the use of “Computational” is too inclusive for some of the provided examples, particularly “networking hardware” and “data storage”. These types of equipment have very minimum “computational” capability as compared to central processing units (CPUs) and graphical processing units (GPUs). A broader term such as “electronic load” would be more appropriate for NERC’s definition, like ERCOT’s adoption of the term “Large Electronic Load,” which extends beyond purely computational activities. Having a uniform definition across industry also facilitates standards and compliance activities. Regardless of the term selected to represent computational loads, it also needs to be clear that Inverter-based Resources such as solar PV, battery energy storage systems (BESS), and modern wind turbines are not included under the term.

Proposed Terminology “Computational Load Entity”

As proposed, “Computational Load Entity means the end-user or the entity that hosts end-users that receives electric power for Computational Load.” We believe it would be beneficial to include the same entity naming structure that is used for Generation and Transmission Owners and Operators (e.g., GO’s/GOP’s). In many cases, data center owners and operators will be separate entities and could have different reliability responsibilities.

There is some uncertainty in the phrase “end-user or the entity that hosts end-users.” At a given facility, if both exist, is the Computational Load Entity inclusive of both of those parties? Or if both exist, is it just the end-user with compliance obligations, and the entity that hosts the end-users has no compliance obligations?

In addition, the use of “end-user” introduces some ambiguities. Traditionally, end-user has been thought of as the entity which uses electricity for a given purpose. End-users could be entities that are lessees of facilities in the computation load facility. Holding an end-user/lessee of computational facilities responsible for compliance seems to be intractable. As end-users/lessees rent computational capabilities, the lessees can frequently change, they could be difficult to identify, and there could be very large numbers of end-users who shouldn’t have any compliance obligations on the BPS.

Appendix 5A

No comments.

Appendix 5B

The proposed criteria is “III.c.1 Computational Load Entity that 1) contributes to an aggregate connected Load capability greater than or equal to 20 MW; 2) at a single point of interconnection to the Bulk Power System at a voltage greater than or equal to 60 kV; and 3) hosts 1 MW or greater of Computational Load.”

With the proposed criteria, there is a risk that certain facilities interconnected at a single point of interconnection greater than 60 kV may be required to register even though they do not pose a risk to the BPS. For example, if a manufacturing facility is 75 MWs, and has 1 MW of Computational Load at the facilities (used in servers for process control, data storage, personal computers, PLC’s, RTU’s, routers, etc.), that facility would need to be registered, even though it does not pose the

same operational characteristics and risks that were identified in the NERC white papers, alerts and guidelines.

The 1 MW of Computational Load compared to 20 MW of aggregate load is 5%. As the aggregate load becomes larger, the percentage of computational load that would be subject to the criteria is reduced, as shown by the example above. Other entities have proposed higher percentages, reflective of the risk on the BPS. For example, ERCOT's proposed definition of Large Electronic Load in NPRR 1308 is "A Large Load in which **50% or greater** of the Demand at the site consists of power electronic based load, specifically computational load, such as data centers and cryptocurrency mining facilities," and the accompanying definition of Large Load is "One or more Facilities at a single site with an aggregate peak Demand greater than or equal to **75 MW** behind one or more common Points of Interconnection (POIs) or Service Delivery Points."

As NERC develops reliability standards for all North America, it should establish minimum acceptable requirements, which would support significantly higher percentage thresholds of computational loads. Having computational loads which are 5% or lower of the total aggregate demand is not reflective of the minimum acceptable level of requirements consistent with some ISOs/RTOs. While we understand that FERC mandated loads greater than 20 MW in their ANOPR, we believe this definition requires further discussion. It is suggested that NERC leverage the LLWG's conclusions to date to develop load criteria aligned with actual reliability risk, while permitting transmission entities to adopt more stringent requirements, if appropriate for their systems.

Thank you for the opportunity to provide these comments.

Tim Taylor

SEIA NERC Working Group Lead

NERC,

The Sacramento Municipal Utility District (SMUD) and the Balancing Authority of Northern California (BANC) support the joint comments submitted by the Trade Associations (APPA, LPPC, EEI, NRECA and TAPS).

Tim Kelley, PE

Senior Regulatory Advisor, Government Affairs

SMUD | Powering forward. Together.

6201 S Street, Mail Stop B404, Sacramento, CA 95817

P.O. Box 15830, Sacramento, CA 95852-0830

Hello,

I would like to submit on behalf of Salt River Project that SRP supports the joint comments submitted by the Trade Associations (APPA, LPPC, EEI, NRECA, and TAPS).

Thank you,

Tim Singh

Principal Engineer

SRP | Regulatory Audit & Compliance | PAB4TE

P.O. Box 52025, Phoenix, AZ 85072-2025

Trade Associations Comments

Thank you for the opportunity to comment on the draft proposed revisions to the North American Electric Reliability Corporation's (NERC) Rules of Procedure (ROP) on the registration of large computational loads. The Trade Associations,¹ on behalf of their members, support NERC's objective of protecting the bulk electric system from reliability risks associated with large computational loads. The Trades understand there are various considerations needed in developing registration criteria and mechanisms for the owners and operators of computational loads.

To achieve our shared objective, the Trade Associations provide these comments in four parts:

- I. **Overarching Principles:** We describe core principles that form the basis for our proposed and alternative solutions in sections II and III, as well as our comments to NERC's proposal in section IV. We urge NERC to consider these principles when drafting the registration criteria for new entities.
- II. **Proposed Solution:** We propose changes to definitions and registration criteria that focus on the highest priority risks.
- III. **Alternative Solution:** We propose alternate definitions and registration criteria that would define two categories of large computational loads.
- IV. **Comments with NERC's Proposal:** If NERC does not accept the Trade Associations' proposed or alternative solutions, we describe the concerns we have with NERC's draft proposal that we urge NERC to address.

I. OVERARCHING PRINCIPLES

DATA CENTER REGISTRATION SHOULD BE RISK-BASED.

Treating different-sized facilities the same would depart from NERC's longstanding risk-based approach to registration. Registration criteria should focus on the loads most likely to have a material impact on bulk electric system (BES) reliability and should avoid imposing unnecessary compliance obligations on facilities that do not present comparable reliability risks.

¹ The Trade Associations are the: American Public Power Association (APPA), Edison Electric Institute (EEL), Large Public Power Council (LPPC), National Rural Electric Cooperative Association (NRECA), Transmission Access Policy Study Group (TAPS), and the National Association of Regulatory Utility Commissioners (NARUC) as a member association.

THE DEFINITION SHOULD CAPTURE LARGE DATA CENTERS AND CRYPTO MINES BUT NOT CAPTURE TRADITIONAL COMMERCIAL OR INDUSTRIAL LOADS.

The registration criteria should focus on the category of large computational loads that NERC has identified as potentially presenting material reliability risks. They should not inadvertently capture traditional commercial or industrial loads that contain information technology equipment but do not operate primarily as computational load facilities, because such loads do not pose material risk to the BES.

For example, large office buildings with server rooms, defense infrastructure (e.g., the Pentagon), and industrial facilities (e.g., aluminum smelters or automobile manufacturing plants) should not be registered merely because they use servers, storage, networking hardware, or other information technology equipment in support of their ordinary operations. The criteria should distinguish facilities whose primary load characteristics raise the reliability concerns identified by NERC from facilities that use information technology equipment only as an incidental part of a broader commercial or industrial process.

RESPONSIBILITY FOR IDENTIFYING NEWLY REGISTERED ENTITIES SHOULD NOT FALL ON EXISTING REGISTERED ENTITIES.

Transmission Owners (TO), Transmission Operators (TOP), Transmission Planners (TP), Distribution Providers (DP), and other existing registered entities do not always have—and may lack legal authority to obtain—the information necessary to determine whether a particular customer meets the criteria for registration as a computational load owner or operator. NERC and the Regional Entities should therefore retain responsibility for identifying large computational loads that may meet the registration criteria.

The registration framework should not impose new, open-ended data collection obligations on existing registered entities or require them to determine whether a customer's facilities meet the proposed computational load criteria. Existing registered entities may have relevant system information, but they should not become the default mechanism for identifying, classifying, or certifying customers for registration.

NEW REGISTRATION CRITERIA SHOULD NOT CREATE COMPLIANCE OBLIGATIONS FOR SMALL DISTRIBUTION UTILITIES OR RESULT IN CHANGES TO THE DEFINITION OF BULK ELECTRIC SYSTEM.

Any new registration criteria for large computational load should not directly or indirectly change the criteria for registering distribution providers, nor should it result in changes to the BES definition. For example, if a large computational load interconnects to a non-BES line, that should not inadvertently convert that line into a BES facility and cause the owner of that facility to become a Transmission Owner.

II. TRADE ASSOCIATIONS' PROPOSED SOLUTION

The Trade Associations propose that NERC define computational load facilities as facilities with at least 75 megawatts (MW) of computational load connected at 100 kilovolts (kV)² or higher, and that NERC register the owners (e.g., the landlords) and operators of the computational load facilities.

CONSISTENT WITH OTHER REGISTRATION CATEGORIES, NERC SHOULD DEFINE FACILITIES AND THEN REGISTER THE OWNERS AND OPERATORS OF THOSE FACILITIES.

A. Add a definition of "Computational Load Facility": A clear definition of a Computational Load *Facility* is essential for Project 2026-02 success. Existing standards that currently address the same types of essential actions being proposed for data centers but are applicable to other comparable equipment (like generators) consistently rely on a *Facility*-type definition. This is consistent with how Generator Owner/Operator and Generating Facility are treated, defined, and included in the registration criteria.

Data center sites can have a significant number of physical and electrical configurations. For example, many data center sites can consist of one or more buildings, with each building having one or more connections to the Bulk Power System (BPS) at the same location. In some cases, data center owners may consider that each building is an individual data center, while in other cases the collection of buildings connected to the grid is viewed as the data center. The risk to the BES comes from the aggregate load behavior of facilities *at a single site*. This was emphasized in NERC's "Characteristics and Risks of Emerging Large Loads" whitepaper that, while specific load types and thresholds to define large loads could not be defined across industry at that time, all agreed that any such definition should include "aggregation of load facilities at a single site behind one or more point(s) of interconnection that can pose reliability risks to the BPS."

We therefore recommend the following definition of "Computational Load Facility" – A collection of electrical equipment located at one or more buildings, structures, or installations that is designed to receive power, has an aggregate connected maximum peak demand of Computational Load greater than or equal to 75 MW, and is served through electric facilities at a voltage level greater than or equal to 100 kV from one or more points of connection.

B. Replace "Computational Load Entity" with "Computational Load Owner" and "Computational Load Operator" or alternatively make changes to "Computational Load Entity."

The Trade Associations recommend that the Computational Load Entity definition be replaced with new definitions that align with the "Owner" and "Operator" Model currently in place in the NERC Rules of Procedure for registration criteria. Today we have registration criteria for TO, TOP, Generator Owner (GO), and Generator Operator (GOP). This structure is in place because for transmission and generation facilities, there are often situations that occur where the owner of the facilities is

² NARUC's position is 115kV or higher as opposed to 100kV or higher.

Trade Association's Comments on Proposed Revisions to NERC Rules of Procedure: Computational Loads

different from the operator of the facilities. This registration criteria set up allows different entities to be responsible only for the activities for which they are contractually obligated as the owner and/or operator. For Computational Load Facilities, these facilities have the same type of structure. There are many cases where the owner is different from the operator.

Therefore, because of the actual structure of the facilities, and to align with existing NERC registration criteria, the Trade Associations recommend replacing the Computational Load Entity definition with Computational Load Owner and Computational Load Operator, with proposed definitions for each included below.

“Computational Load Owner” means the end-user entity that owns and maintains a Computational Load Facility, **and/or** the entity that owns and maintains a Computational Load Facility that hosts end-users that receives electric power **at a Computational Load Facility**.

“Computational Load Operator” means the end-user entity that operates a Computational Load Facility, **and/or** the entity that operates a Computational Load Facility that hosts end-users that receives electric power **at a Computational Load Facility**.

However, as discussed in more detail below, if NERC staff retains only one registered entity, the Trade Associations suggest the following modifications to the Computational Load Entity definition:

“Computational Load Entity” means the end-user **and/or** the entity that hosts end-users that receives electric power ~~for~~ **at a Computational Load Facility**.

The risk to the BPS and the gaps in NERC's reliability standards, as noted in NERC's "Assessment of Gaps in Existing Practices, Requirements, and Reliability Standards for Emerging Large Loads" are driven by computational load impacts and behavior, not *other type* of loads, so the Computational Load Entity definition should focus only on the amount of data-centric/information technology type of load.

COMPUTATIONAL LOADS SHOULD BE DEFINED AS LOADS MADE UP PRIMARILY OF POWER ELECTRONICS-BASED INFORMATION TECHNOLOGY EQUIPMENT.

Clarify the Computational Load definition: The Trade Associations recommend that the proposed Computational Load definition focus on the power electronic-based information technology equipment, which is the focus of the risks to the BPS as documented by NERC's Large Loads Task Force and Large Loads Working Group.

NERC's "Assessment of Gaps in Existing Practices, Requirements, and Reliability Standards for Emerging Large Loads" identifies risks associated with the behavior and impacts of computational loads. Those risks arise from the data-centric and information technology characteristics of the load, not from unrelated load at the same site. The definition should therefore focus only on the amount of computational load and should not sweep in other types of loads that do not raise the same reliability concerns.

Therefore, we urge NERC to update it as follows: **“Computational Load”** means Load comprised **primarily** of electric power demand from **power electronic-based** information technology equipment.

THE REGISTRATION THRESHOLD SHOULD BE SET TO 75 MW/100 kV.

The Trade Associations recommend that NERC revise the proposed thresholds in Appendix 5B. The proposed 20 MW/60 kV threshold with the 1MW Computational Load criteria is set unnecessarily low and sweeps in a substantial number of facilities that pose no material impact to the reliability of the BPS and, thus, does not justify the burden of NERC registration. The current Rules of Procedure, Registration, and Standard Drafting efforts related to Computational Loads should focus on those loads that pose the most meaningful risk to the BPS. The threshold criteria (20 MW, 60 kV) is too low and we recommend the threshold be set to 75 MW connected at ≥ 100 kV for Computational Load Facilities, consistent with the NERC BES definition for dispersed power producing resources (i.e., inverter-based resources) – where the current criteria is greater than 75 mega volt amperes (MVA) at a common point of connection at ≥ 100 kV is utilized.

Although NERC has developed a Technical Reasoning Proposed Computational Load document, it does not provide sufficient explanation or practical examples to justify the selected thresholds in the proposed registration criteria. The Trade Associations are concerned that NERC's proposed approach could inadvertently create compliance obligations for DP entities by effectively deeming them TO or TOP. It is critical to ensure that DP do not lose the radial system exemption when solely serving load and not operating transmission facilities. Clear justification and refinement of the proposed thresholds are required to demonstrate that there are identifiable and material risks to the BES.

In addition, NERC's "Characteristics and Risks of Emerging Large Loads" indicated that "most of the survey respondents qualified 'large' as greater than 50 MW, and the single size number most commonly suggested was 75 MW," and the NERC Alert 2 results confirmed this. NERC's "Assessment of Gaps in Existing Practices, Requirements, and Reliability Standards for Emerging Large Loads" references the first Whitepaper and in the definition section, and focuses on really large loads and how several smaller loads (still much bigger than 20 MW) are not as big of a risk. "The potential impact of several 100 MW loads may be less than the impact of a 500 MW load and may require different treatment."

Furthermore, NERC's "Ten benchmarks of an Effective Reliability Standard" white paper emphasizes that "each [NERC reliability standard] requirement is 'not a lowest common denominator' exercise but instead achieves an objective that is the best approach for bulk power system reliability." The same applies to definitions and reliability criteria. NERC's decision to select 20 MW appears to be a "lowest common denominator" choice, instead of a higher value that the majority of the industry agrees with.

The Trade Associations recommend NERC adopt an initial threshold of 75 MW at a common point of connection of 100 kV or higher. This pairing adheres to the current BES definition and appropriately balances reliability risk to compliance burden. A 75 MW/100 kV starting point focuses computational load entity registration on the facilities most likely to have a material impact on BPS reliability. It would also give NERC and the Regional Entities time to identify entities for registration, develop reliability standards, and gain compliance oversight experience with a manageable population. Further, it avoids imposing new obligations on facilities whose individual contribution to reliability risk is limited.

III. TRADE ASSOCIATIONS' ALTERNATIVE PROPOSAL

As an alternative, the Trade Associations recommend that NERC adopt an approach that mirrors the Category 1 and Category 2 Generator Owner and Generator Operator definitions adopted as part of NERC's Inverter-Based Resources Registration Initiative.

THE NEW REGISTRATION FRAMEWORK SHOULD STILL DEFINE FACILITIES AND THEN REGISTER THE OWNERS AND OPERATORS OF THOSE FACILITIES.

As with our primary proposal, our alternative proposal retains the structure of defining Computational Load Facilities and registering the owners and operators of those facilities.

THE NEW REGISTRATION SHOULD CONSIST OF TWO CATEGORIES OF COMPUTATIONAL LOAD OWNERS AND OPERATORS.

If a threshold lower than 75 MW is deemed necessary by NERC staff, the Trade Associations recommend that NERC staff enhance the Technical Reasoning document to reflect the analysis/evidence demonstrating the lower facility threshold is needed for BPS reliability. In addition, in this scenario, the Trade Associations also recommend that NERC staff adopt the same registration criteria model that was developed for inverter-based resources with the Category 1 GO/GOP and Category 2 GO/GOP. NERC should create a new Category 1 Computational Load Owner/Operator (CLO/CLOP) and Category 2 CLO/CLOP definitions, and structure the MW and voltage criteria for each to align with the Category 1 GO/GOP and Category 2 GO/GOP.

With NERC expected to address computational load risk on an accelerated timeline by the end of 2026, this structure would allow ERO Enterprises to focus its initial entity registration, standards development efforts, and compliance education activities on the highest-risk CLOs/CLOPs first. This framework also establishes a more manageable path to subsequently address lower-risk Category 2 CLOs/CLOPs in future work phases. Finally, this structure would enable a risk-based compliance approach by allowing more stringent reliability and security requirements to be applied to Category 1 CLOs/CLOPs compared to Category 2 CLOs/CLOPs.

Under this alternative structure, the following definitions are proposed:

“Computational Load Facility” means a collection of electrical equipment located at one or more buildings, structures, or installations that is designed to receive power, is served through electric facilities from one or more points of connection **and meets the criteria in either Category 1 or Category 2 Computational Load Owner/Operator registration categories.**

“Category 1 CLO/CLOP” should be aligned with the BES criteria, which would be entities who own/operate Computational Load Facilities (CLFs) that are **>= 75 MW** of Computational Load connected at **>= 100kV**.

“Category 2 CLO/CLOP” should be aligned with the lower threshold value supported by NERC and industry, which would be entities who own/operator Computational Load Facilities that are **>= 20 MW** of Computational Load connected at **>= 60kV**.

IV. COMMENTS ON NERC'S PROPOSED REVISIONS

As noted earlier, the Trade Associations have several concerns with NERC's proposed revisions to the Rules of Procedure. If, however, NERC declines to adopt either solution, and proceeds with the approach in the April 1, 2026, draft, we urge NERC to address the following concerns before finalizing any revisions.

REPLACE “CAPABILITY” WITH “MAXIMUM PEAK DEMAND.”

The draft Computational Load Entity registry criteria use the word “capability,” which can be interpreted as the maximum load of a Computational Load Facility (CLF). For example, if the CLF facility has a maximum possible load demand of 100 MW, then this is the load “capability” of the CLF. However, this “capability” term as used with load is likely not widely understood or utilized today.

We recommend an alternative to the use of “capability” such as **“maximum peak demand”** of a facility to be clearer. In addition, evaluating a CLF's impact on the BES should not rely solely on its size. Incorporating operational characteristics, such as interruptibility, load flexibility, and the availability of on-site or contracted backup generation, will provide a more comprehensive and accurate assessment of system impacts.

CLARIFY HOW “POINT OF INTERCONNECTION” APPLIES TO LARGE COMPUTATIONAL LOADS.

The proposed registration criteria include a reference to a “point of interconnection” at 60 kV. “Point of interconnection” is primarily a term referring to generation interconnections – interconnections that most often rely on one generating tie-line. Data centers often have many connections to the BPS at the same location. The ownership of equipment can also vary site to site, in which some connect to utilities directly at the high voltage kV bus by owning the high-voltage transformation, while others do not own the transformation or equipment greater than 60 kV.

CLARIFY HOW TOTAL AGGREGATE LOAD WILL BE CALCULATED.

NERC's current proposed Computational Load Entity registration criteria would register an entity that "contributes to an aggregate connected Load capability" of 20MW or greater. How will NERC determine the amount of Computational Load or the amount of aggregate connected load thresholds? Will it be based on the "nameplate" of the various racks in a data center warehouse or some other methodology? Regardless of the method(s) for determining these values, the Trade Associations recommend that this should be confirmed between the CLE and its Regional Entity and NERC should not be dependent on other registered entities for providing this information to Regional Entities. Critically this criteria state load, not Computational Load. The Trade Associations believe this is incorrect and should be updated to state "Computational Load capability." In our proposed Computational Load Facility definition, we have included the use of Computational Load in the definition so that the focus is only on data-centric/information technology load.

NERC should also clarify how it will calculate the relevant load quantity. For example, NERC should explain whether the calculation will rely on the nameplate rating of server racks or other equipment, contract demand, projected maximum peak demand, measured demand, or another methodology. Clear rules will reduce uncertainty and improve consistency across regions.

THE 1 MW THRESHOLD SHOULD BE ELIMINATED.

It is unclear how, in NERC's April 1 proposal, a Computational Load is fundamentally different from other large traditional loads, such as steel mills, food processors, or automobile manufacturers. All large loads must be studied to ensure their interconnection and operation do not create grid reliability issues. There is concern that, as written, it may capture facilities that are not data centers or cryptocurrency mining operations, which are the business types that NERC has identified as potentially impacting the reliability of the BES. For example, many industrial and commercial facilities may operate more than 1 MW of information technology equipment (either now or in the future), such as servers, storage, and networking hardware to control and support processes, yet these are not solely or primarily computational in nature. In addition, it seems unreasonable that the loss of a 1 MW load, regardless of load type, would have a significant impact on the BES, therefore including a de minimis criteria of 1 MW of Computational Load drives significant confusion, pulls in a significant number of entities unnecessarily, and has no justification pertaining to actual risk to the BPS. The 1 MW de minimis threshold conflicts with the material impact standard applied throughout Appendix 5B and would expand registration beyond entities capable of materially impacting BPS reliability. The 1 MW de minimis threshold should be removed from the Computational Load threshold.

THE TERM "ENTITIES THAT HOST END-USERS" IS UNCLEAR AND SHOULD BE REPLACED.

The proposed materials use the phrase "entities that host end-users." That phrase appears intended to address data center owners or landlords that host tenants. But the phrase could also be read more broadly.

Trade Association's Comments on Proposed Revisions to NERC Rules of Procedure: Computational Loads

For example, a distribution utility may own a substation that serves 20 MW of load, including a credit card processing facility with 2 MW of servers. In that circumstance, the distribution utility could be described as “hosting” the load on its system, even though it does not own or operate the computational load facility. The registration criteria should not create that result.

NERC should replace or clarify the phrase “entities that host end-users” to ensure that it does not include distribution utilities, retail service utilities, TO, TOP, TP, or DP that merely serve, study, or interconnect computational load facilities. The registered entity should be the entity that owns or operates the Computational Load Facility, not the utility that provides electric service to it.

As noted above, the Trades suggest the following language to address the concern:

“Computational Load Entity” means the end-user **and/or** the entity that hosts end-users that receives electric power **for at a** Computational Load **Facility**.

ADDITIONAL TRADE ASSOCIATIONS QUESTIONS REGARDING NERC'S APRIL 1 PROPOSAL

To build on the concerns shared and the alternatives provided the Trade Associations ask the following questions

- “How will a Computational Load Entity(ies) be identified for registration under the proposed framework?”
- Will the TO, TOP, or TP be responsible for identifying this/these load(s) and sharing that information with their Regional Entity?
- Additionally, will responses to the NERC Level 3 Alert on Computational Loads be used to inform or determine registration requirements?

In addressing these questions, the Trade Associations recommend that the identification CLEs for potential registration should be the responsibility of the Regional Entity and NERC should not be dependent on other registered entities for providing this information to Regional Entities.

CONSIDER AN EXCEPTION PROCESS FOR LARGE COMPUTATIONAL LOADS THAT MEET THE CRITERIA BUT CAN BE SHOWN NOT TO HAVE A MATERIAL IMPACT ON THE BULK ELECTRIC SYSTEM.

Brightline Criteria

Computational Load Entity (CLE) registration should be based on clear, brightline criteria coupled with a formal exception process similar to the BES definition framework. Such an approach would allow an entity to demonstrate, through documented technical justification, that its facilities do not have a material impact on the BES. This balance is necessary to ensure that registration and compliance obligations are applied only where there is a substantiated reliability risk, while

Trade Association's Comments on Proposed Revisions
to NERC Rules of Procedure: Computational Loads

avoiding unnecessary regulatory burden for entities whose operations do not pose a material threat to BES reliability.

NERC SHOULD DEVELOP A REFERENCE DOCUMENT.

The proposed criteria raise practical questions about how facility topology and interconnection configurations will affect registration. NERC should develop a reference document addressing the new registration criteria. The document should give industry and the drafting team a clear opportunity to review how NERC and the Regional Entities expect to apply the criteria. That reference document would promote consistent implementation and reduce the risk of unintended consequences.

CANDIDATES MUST MEET ALL CRITERIA TO BE REGISTERED.

The Trade Associations also note that NERC's document "Summary of Revisions" states candidates for Computational Load registration must meet all the criteria in the proposed revisions in Appendix 5B to be registered as a Computational Load Entity; however, that is not explicitly stated in the proposed changes in Appendix 5B and should be stated.

Submitted Electronically to ROPComments@nerc.com

May 22, 2026

Marisa Hecht
Senior Counsel, Legal and Regulatory
North American Electric Reliability Corporation

Re: American Chemistry Council Comments on Proposed NERC Computational Load Entity Rules of Procedures Revisions

Dear Ms. Hecht:

The American Chemistry Council (ACC)¹ appreciates the opportunity to provide input on the proposed revisions to the North American Energy Reliability Corporation (NERC) Rules of Procedures (ROP) to create a new category of registered customer entities, “Computational Load Entities,” and associated reliability considerations on the Bulk Power System (BPS). ACC and our members recognize the essential role NERC plays in ensuring the reliability of America's electricity grid and chemical sector's continuous access to affordable and reliable industrial power. NERC's leadership is more important than ever during this pivotal period of rising demand, evolving load and generation characteristics, and the need for rapid deployment of critical energy infrastructure.

ACC represents a diverse set of companies engaged in the business of chemistry, a \$673 billion enterprise. The chemical manufacturing industry is an energy intensive and trade exposed cornerstone of American competitiveness, national security, and technological innovation. ACC's members produce a wide variety of chemicals, polymers, and related products that make modern life safer, more productive, and more sustainable, supporting over 25 percent of U.S. GDP and directly touching nearly all manufactured goods. As a foundational sector of the U.S. economy, the chemical industry plays a critical role in enabling domestic manufacturing, supply chain resilience, and long-term economic growth.

¹ The American Chemistry Council's mission is to advocate for the people, policy, and products of chemistry that make the United States the global leader in innovation and manufacturing. To achieve this, we: Champion science-based policy solutions across all levels of government; Drive continuous performance improvement to protect employees and communities through Responsible Care®; Foster the development of sustainability practices throughout ACC member companies; and Communicate authentically with communities about challenges and solutions for a safer, healthier and more sustainable way of life. Our vision is a world made better by chemistry, where people live happier, healthier, and more prosperous lives, safely and sustainably—for generations to come.

While ACC's membership is diverse, our members share one common characteristic: heavy reliance on reliable, around the clock, affordable energy inputs to drive operations and thermally intensive processes and, in many cases, to serve as essential chemical feedstocks. Chemical manufacturing accounts for approximately 40 percent of industrial sector energy consumption, including 20 percent of electricity, 42 percent of natural gas, and 56 percent of purchased steam. Over the past 15 years, the U.S. chemical sector has become a global leader in manufacturing competitiveness, driven in significant part by access to abundant and reliable energy supplies, including electricity.

As NERC, its regional partners, and states address the infrastructure and reliability foundations needed to support hyperscale computational load and other sources of demand growth, ACC urges careful consideration of how resulting policies could intentionally or unintentionally affect the competitiveness of American industrial customers and lead to outcomes that put customers at greater financial and compliance risk without achieving commensurate reliability protections.

ACC offers the following recommendations, beginning with considerations that would be essential under any potential path forward, followed by concerns regarding process and statutory authority, with a focus on avoiding unintended expansion of regulatory scope and protecting energy-intensive industrial customers from disproportionate impacts.

Should NERC elect to proceed with regulating electricity customers in any manner, such policy should incorporate clear guardrails and limitations. Definitions and registration criteria must be narrowly and precisely drawn to avoid inadvertent overlap with traditional industrial and manufacturing loads and incorporate an explicit limiting principle confirming that the action is not intended to establish a precedent or pathway for future expansion of NERC authority to other end use customer groups. Absent such clarity and constraints, the risk of unintended scope creep, regulatory uncertainty, and collateral impacts on energy intensive industries that may not present the same reliability considerations for the BPS would be substantial.

More fundamentally, ACC urges NERC to reconsider the premise of the proposed Computational Load Entity ROP Revisions, which would expand mandatory reliability registration and compliance obligations to end use customer groups. Such an expansion would be unprecedented, raising serious questions regarding the scope and intended reach of Section 215 of the Federal Power Act. Extending NERC's regulatory scope in this manner would introduce a new layer of regulatory burden by shifting or diffusing responsibility away from existing registered entities—those with operational control of the bulk power system, established protocols, and the technical workforce necessary to execute reliability obligations—toward users that generally lack the specialized expertise, staffing, systems visibility, and compliance infrastructure to assume grid reliability functions.

Finally, given the novel expansion of NERC's regulatory scope, its implications for the scope of NERC's authority under Section 215 of the Federal Power Act, and the resulting application to industries not previously subject to NERC oversight, ACC would recommend, NERC expand its

formal public consultation process and conduct a robust regulatory and economic impact assessment, consistent with the level of scrutiny that would accompany any other significant federal regulatory action. That process should include broad outreach to both computational load customers and other large-load industries potentially affected by future efforts to register end-use customers. ACC also recommends that NERC's impact assessment clearly demonstrates why the reliability concerns identified cannot be addressed through refinements to existing standards and obligations applicable to entities already subject to NERC oversight and should establish this proceeding as an exception rather than a general model for managing grid reliability.

ACC appreciates the opportunity to comment and looks forward to continued engagement as NERC adapts and updates its policies to address today's grid challenges. We would welcome the opportunity to consult further should the drafting team have any questions. Please feel free to contact me at Charles_Franklin@americanchemistry.com or (202) 249 6412.

Sincerely,

Charles Franklin

Charles Franklin
Senior Director, Energy, Climate and Environment
Regulatory and Scientific Affairs
American Chemistry Council